

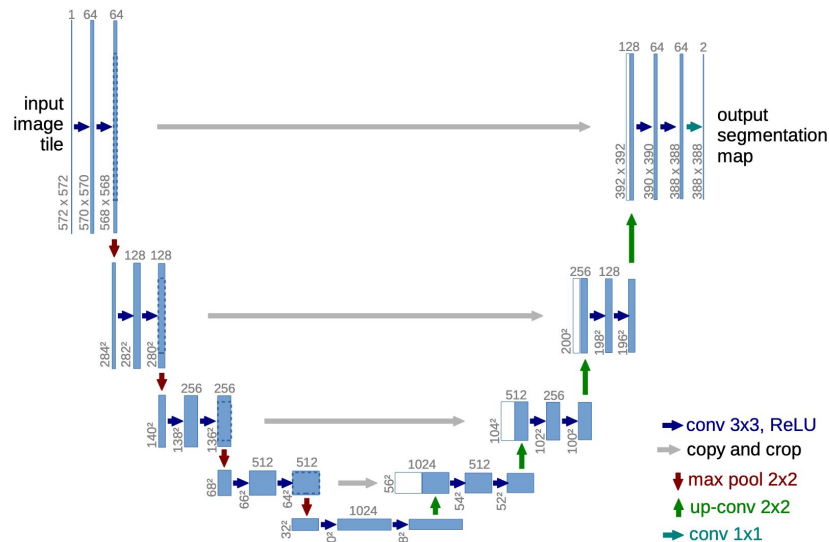
CME 296: Diffusion & Large Vision Models



Afshine Amidi & Shervine Amidi



Recap of last episode...

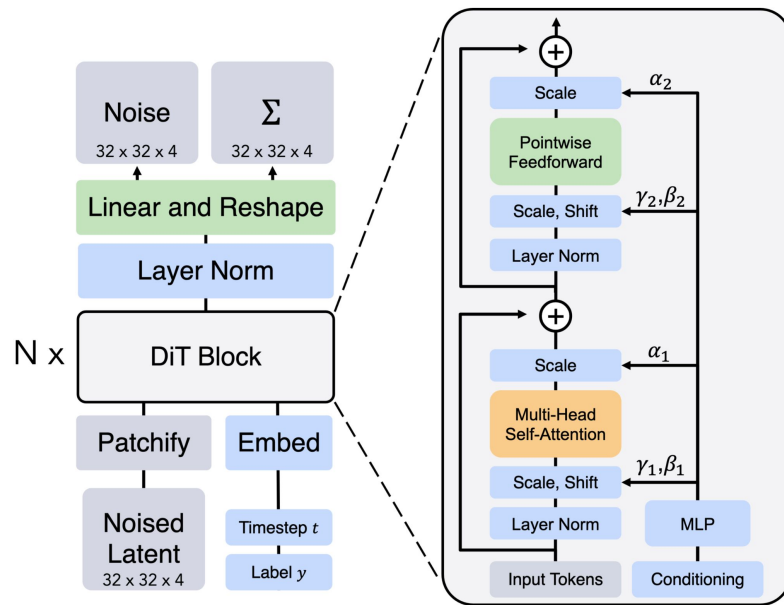


U-Net

Recap of last episode...

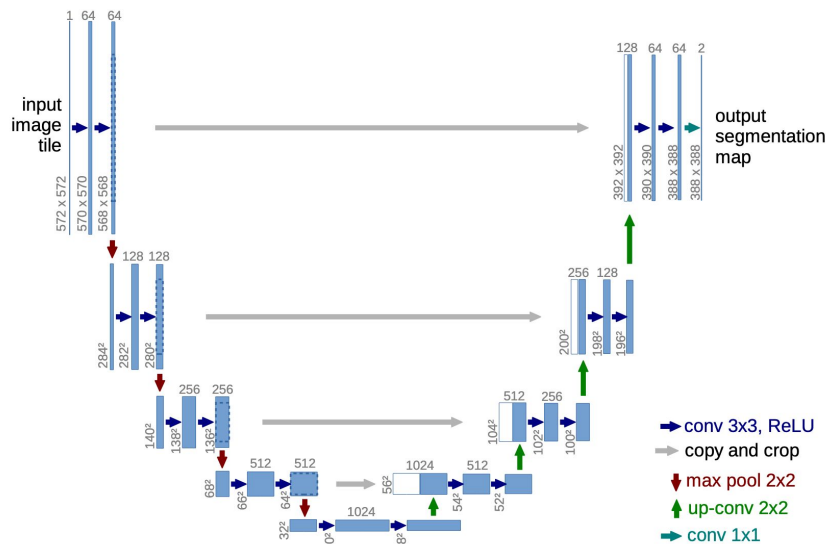


U-Net

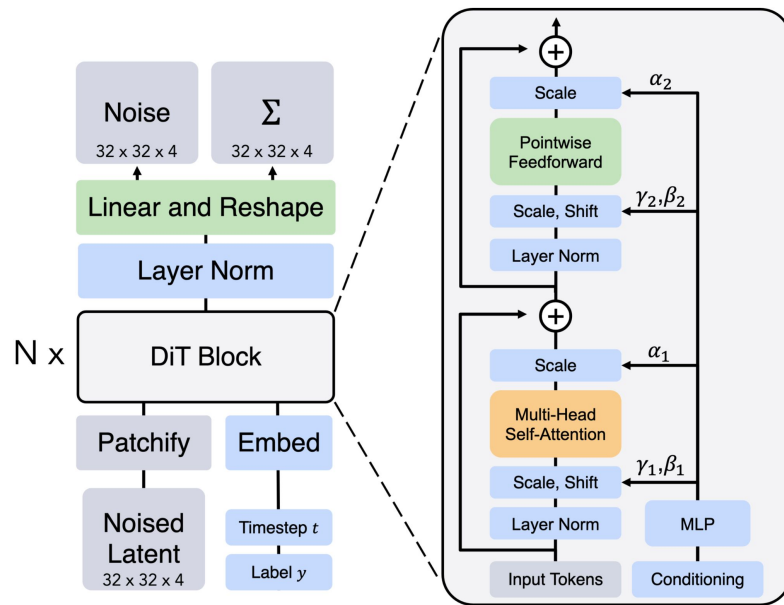


Diffusion Transformer

Recap of last episode...

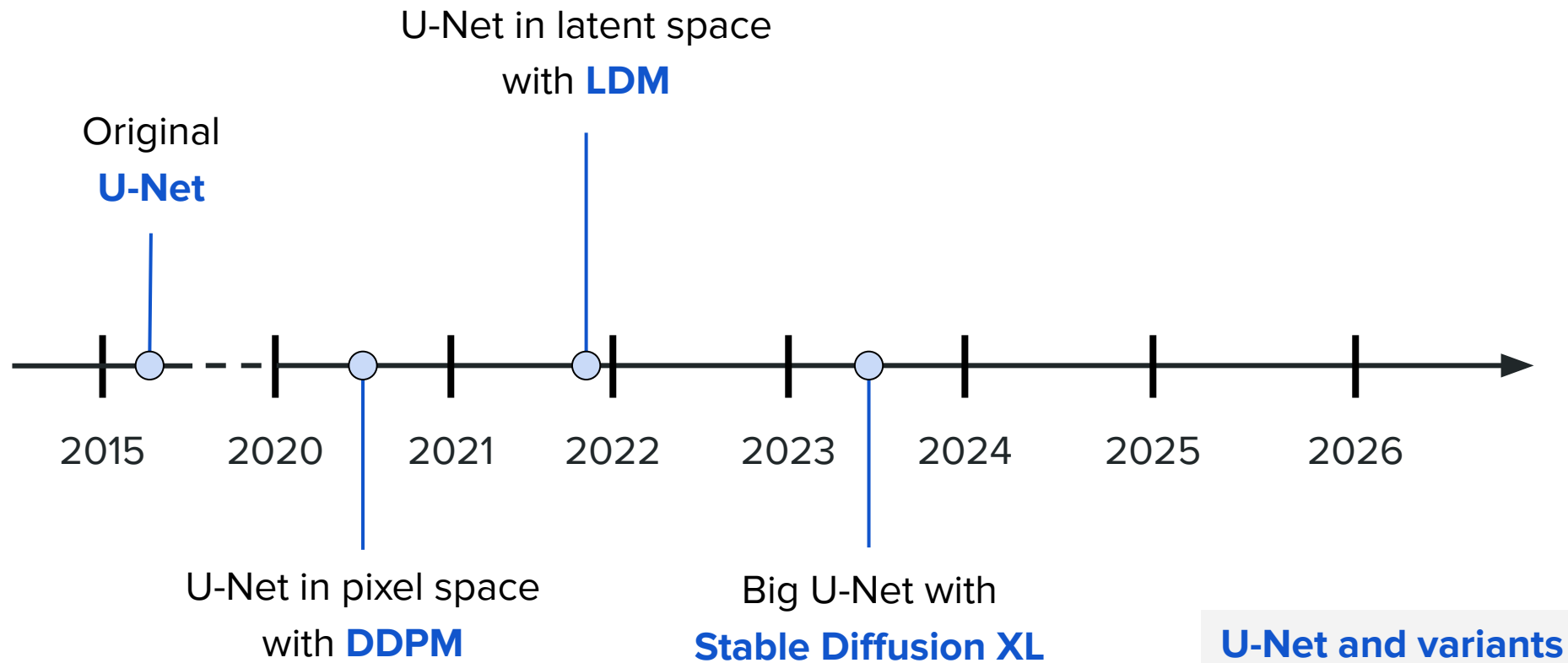


U-Net
& variants

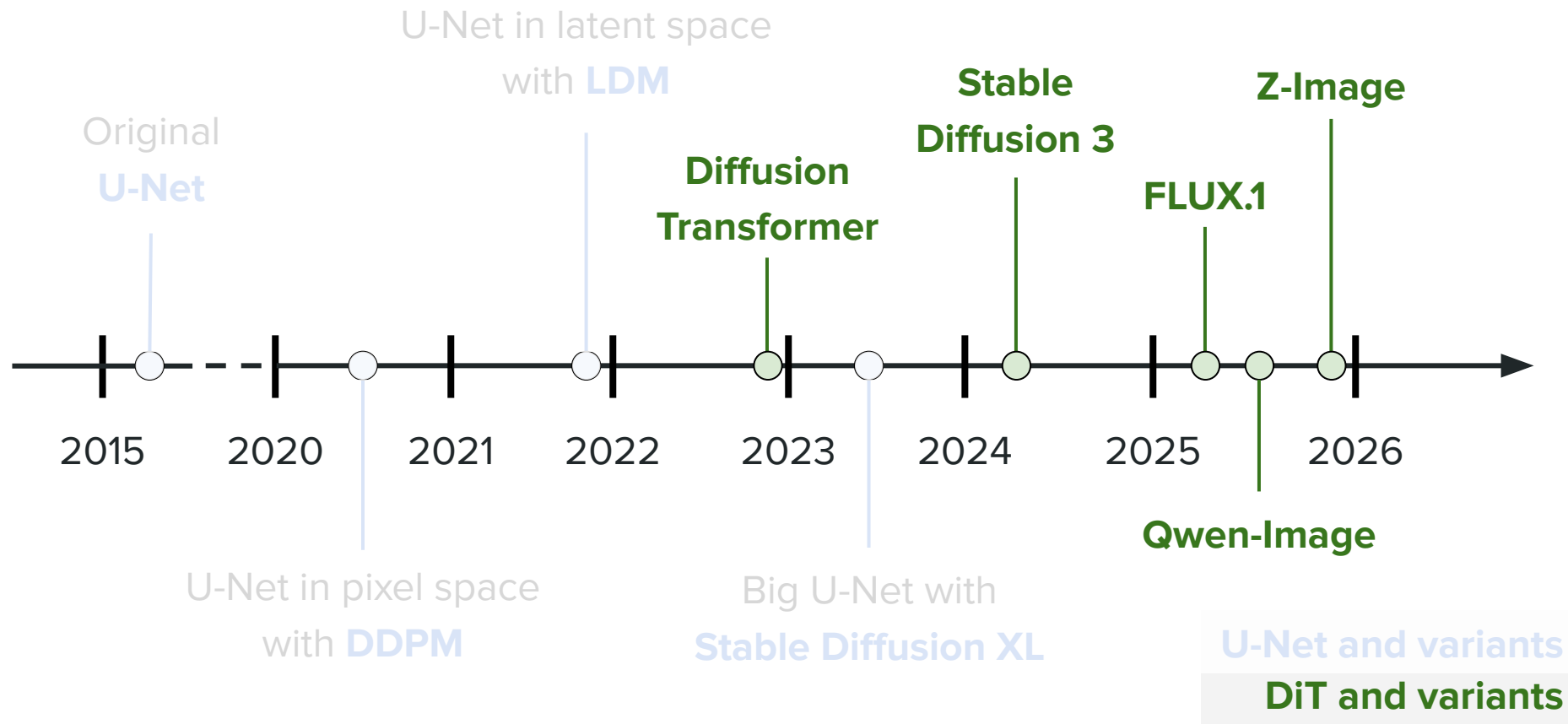


Diffusion Transformer
& variants

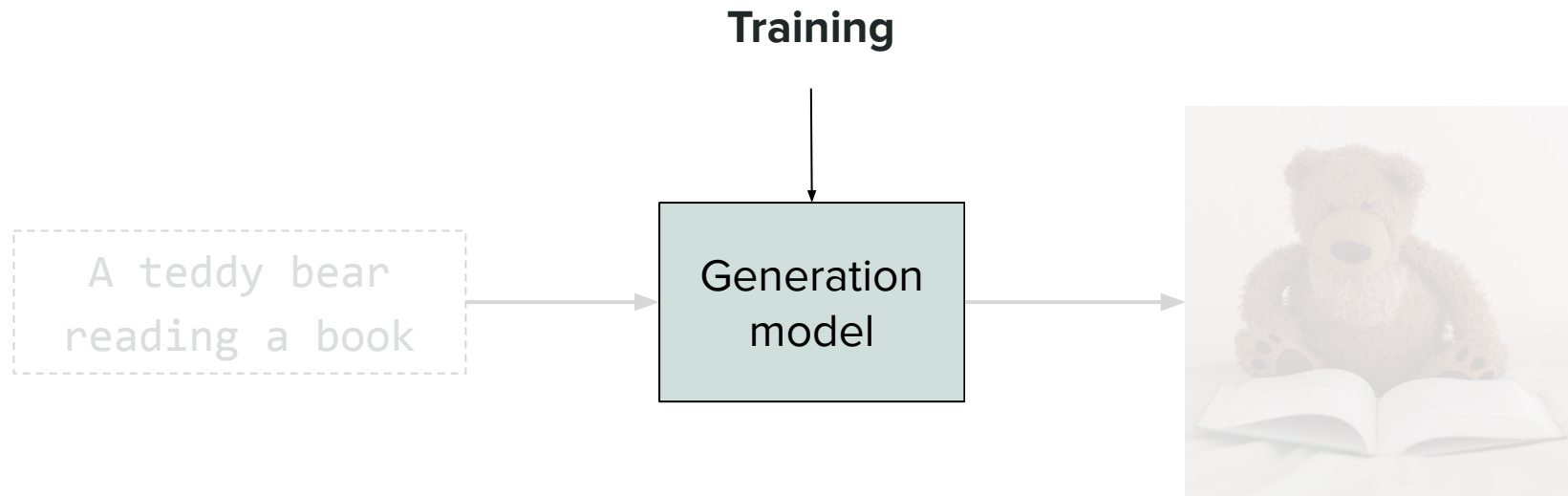
Recap of last episode...



Recap of last episode...



Today's lecture



Today's lecture: **Training lifecycle**

Scope

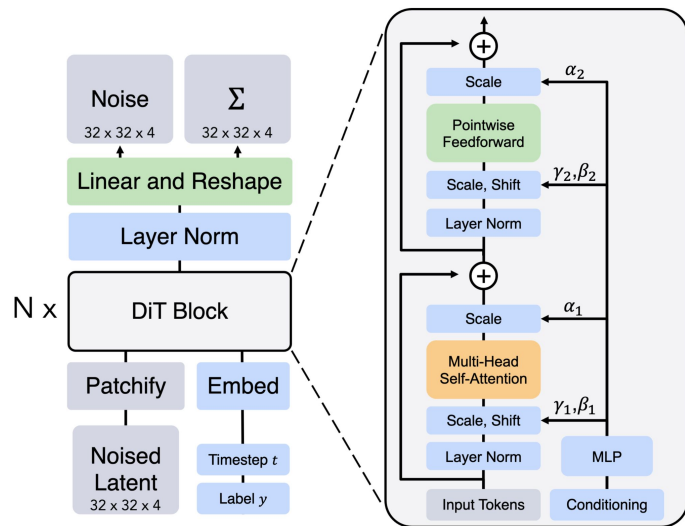
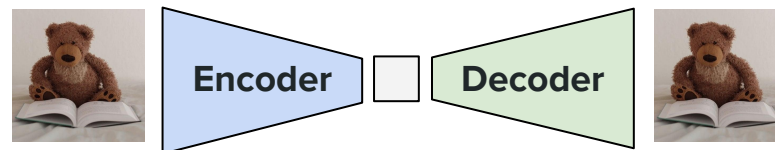
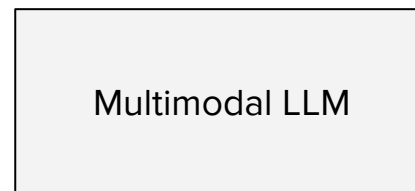


Image generation model



Variational AutoEncoder



Embedding models

Scope

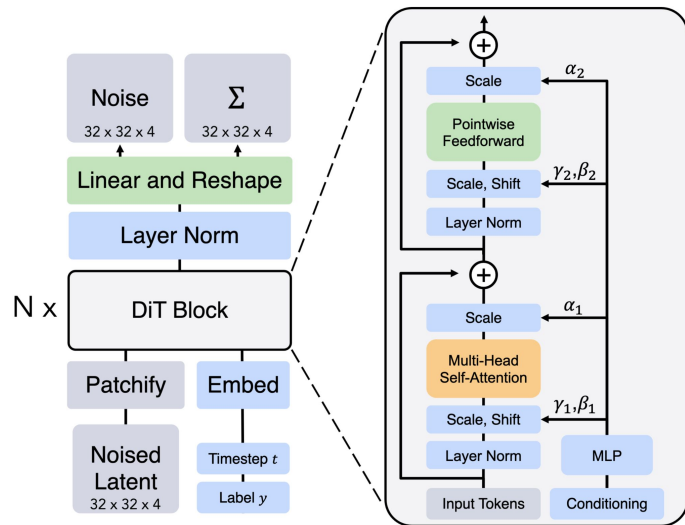


Image generation model

Variational AutoEncoder

Embedding models



Diffusion & Large Vision Models

Lifecycle overview

Training loss

Pre-training

Post-training

Tuning

Distillation

Lifecycle

1

Pre-training



Learn **how** to
generate images

Lifecycle

1

Pre-training



Learn **how** to
generate images

2

Post-training



Learn how to
generate **good**
images

Lifecycle

1

Pre-training



Learn **how** to
generate images

2

Post-training



Learn how to
generate **good**
images

3

Tuning



Learn how to
generate images
for a special case

Lifecycle

1

Pre-training



Learn **how** to
generate images

2

Post-training



Learn how to
generate **good**
images

3

Tuning



Learn how to
generate images
for a special case

4

Distillation



Learn how to
generate images,
fast



Diffusion & Large Vision Models

Lifecycle overview

Training loss

Pre-training

Post-training

Tuning

Distillation

Loss parameterization

**Diffusion
with DDPM**

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) \right) + \sigma_t \epsilon$$

$$\epsilon_{\theta}(x_t, t)$$

Loss parameterization

**Diffusion
with DDPM**

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2 \right]$$

Loss parameterization

Diffusion
with DDPM

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2 \right]$$

Score
matching

$$dx = f(x_t, t)dt + g(t)dW$$

$$dx = \left[f(x_t, t) - g(t)^2 s_{\theta}(x_t, t) \right] dt + g(t)d\bar{W}$$

$$s_{\theta}(x_t, t)$$

Loss parameterization

Diffusion
with DDPM

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2 \right]$$

Score
matching

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t, x_0, x_t} \left[\lambda_t \|s_{\theta}(x_t, t) - \nabla_{x_t} \log p(x_t \mid x_0)\|^2 \right]$$

Loss parameterization

**Diffusion
with DDPM**

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2 \right]$$

**Score
matching**

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t, x_0, x_t} \left[\lambda_t \|s_{\theta}(x_t, t) - \nabla_{x_t} \log p(x_t \mid x_0)\|^2 \right]$$

**Flow
matching**

$$x_t = (1 - t)x_0 + tx_1$$

$$x_{t+\Delta t} = x_t + u_{\theta}(x_t, t)\Delta t$$

$$u_{\theta}(x_t, t)$$

Loss parameterization

**Diffusion
with DDPM**

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2 \right]$$

**Score
matching**

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t, x_0, x_t} \left[\lambda_t \|s_{\theta}(x_t, t) - \nabla_{x_t} \log p(x_t \mid x_0)\|^2 \right]$$

**Flow
matching**

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_1, x_0} \left[\|u_{\theta}(x_t, t) - (x_1 - x_0)\|^2 \right]$$

Loss parameterization

**Diffusion
with DDPM**

$$\mathcal{L}_{\text{DDPM}} = \mathbb{E}_{t, x_0, \epsilon} \left[\|\epsilon_{\theta}(x_t, t) - \epsilon\|^2 \right]$$

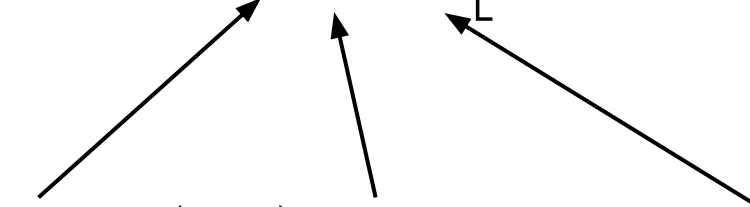
**Score
matching**

$$\mathcal{L}_{\text{DSM}} = \mathbb{E}_{t, x_0, x_t} \left[\lambda_t \|s_{\theta}(x_t, t) - \nabla_{x_t} \log p(x_t \mid x_0)\|^2 \right]$$

**Flow
matching**

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_1, x_0} \left[\|u_{\theta}(x_t, t) - (x_1 - x_0)\|^2 \right]$$

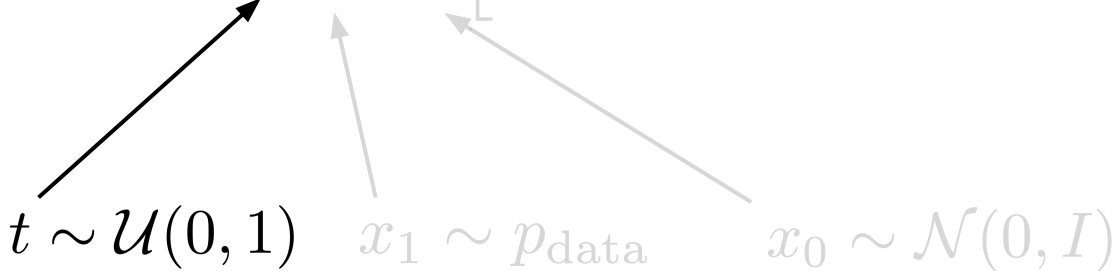
Objective

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_1, x_0} \left[\|u_\theta(x_t, t) - (x_1 - x_0)\|^2 \right]$$


The diagram illustrates the components of the expectation operator $\mathbb{E}_{t, x_1, x_0}$ in the loss function. Three arrows point from the sampling equations below to the variables t , x_1 , and x_0 in the expectation operator:

- $t \sim \mathcal{U}(0, 1)$ (Uniform distribution)
- $x_1 \sim p_{\text{data}}$ (Data distribution)
- $x_0 \sim \mathcal{N}(0, I)$ (Gaussian distribution)

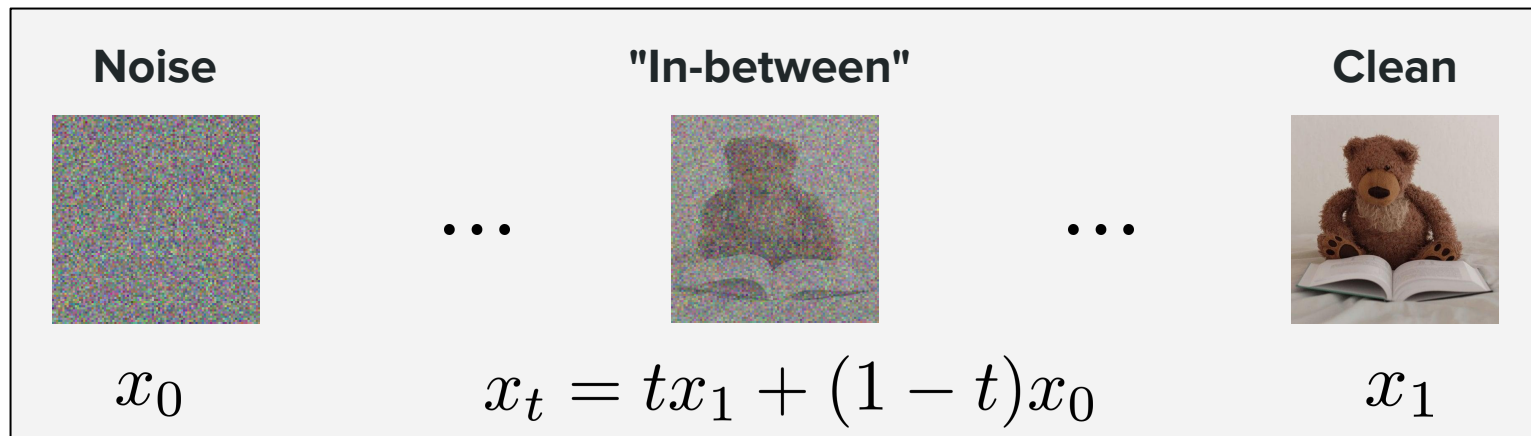
Objective

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_1, x_0} \left[\|u_\theta(x_t, t) - (x_1 - x_0)\|^2 \right]$$


$t \sim \mathcal{U}(0, 1)$ $x_1 \sim p_{\text{data}}$ $x_0 \sim \mathcal{N}(0, I)$

Objective

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t, x_1, x_0} \left[\|u_{\theta}(x_t, t) - (x_1 - x_0)\|^2 \right]$$

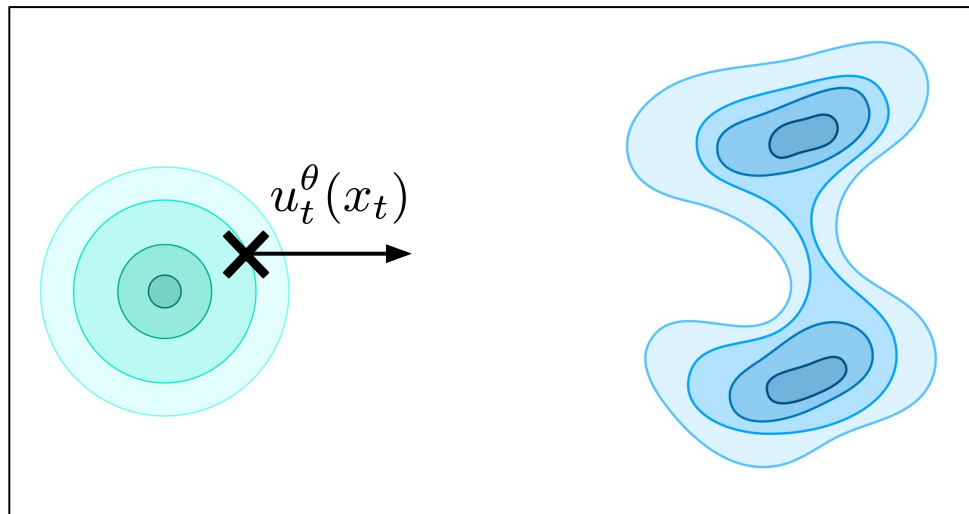


Discussion on task difficulty as a function of t

t close to 0 = **Easy** task



Input x_t

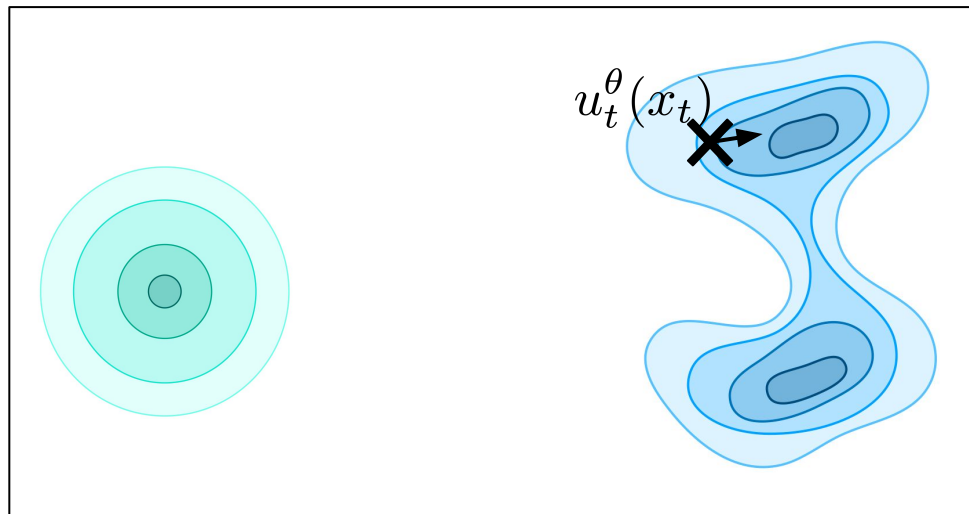


Discussion on task difficulty as a function of t

t close to 1 = **Easy** task



Input x_t

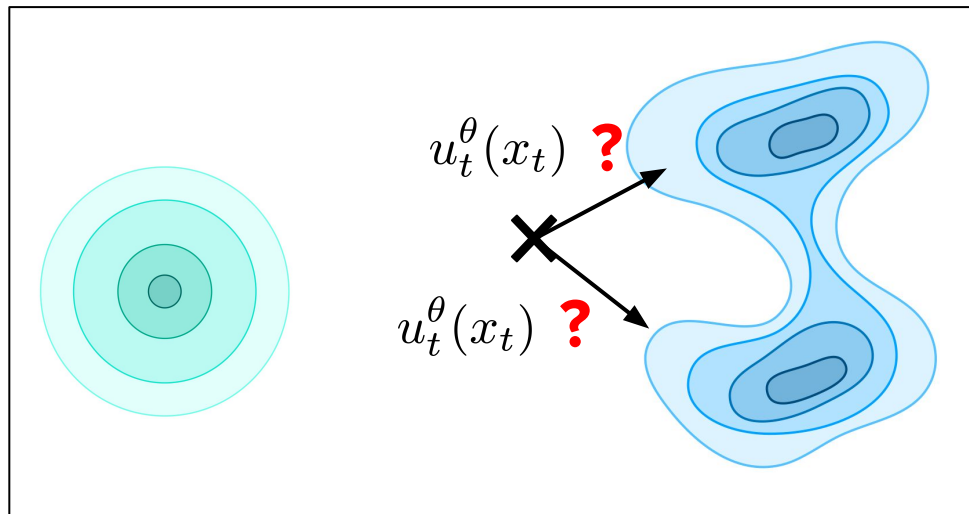


Discussion on task difficulty as a function of t

t around 0.5 = **Hard** task



Input x_t



Proposed distribution for timesteps

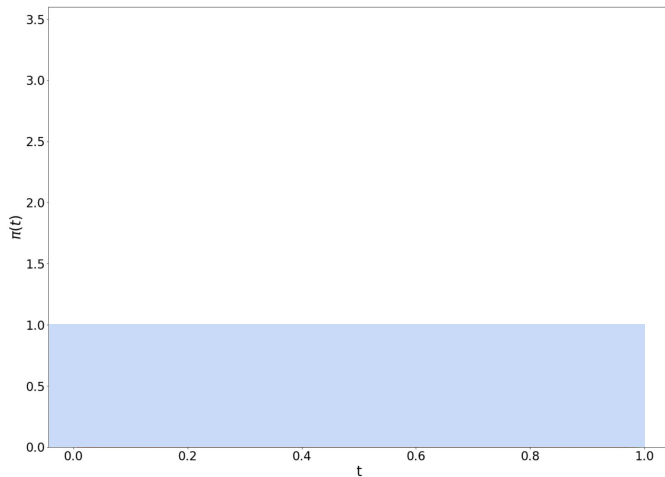
Goal. Emphasize on "hard tasks" at training time

Proposed distribution for timesteps

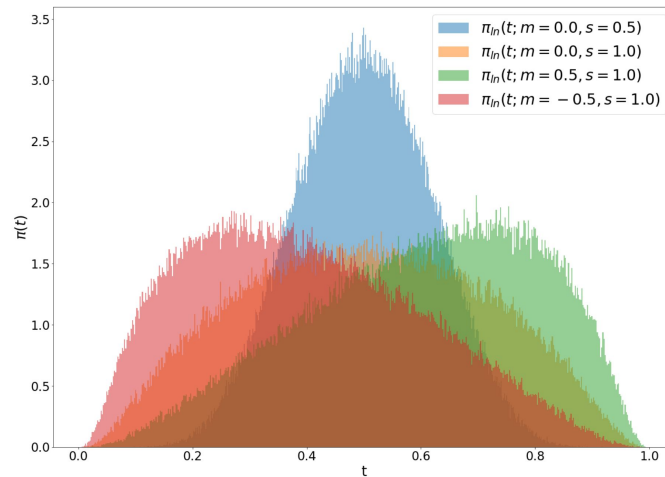
Goal. Emphasize on "hard tasks" at training time

Idea. Use different distribution for timesteps

Before



After



Proposed distribution for timesteps

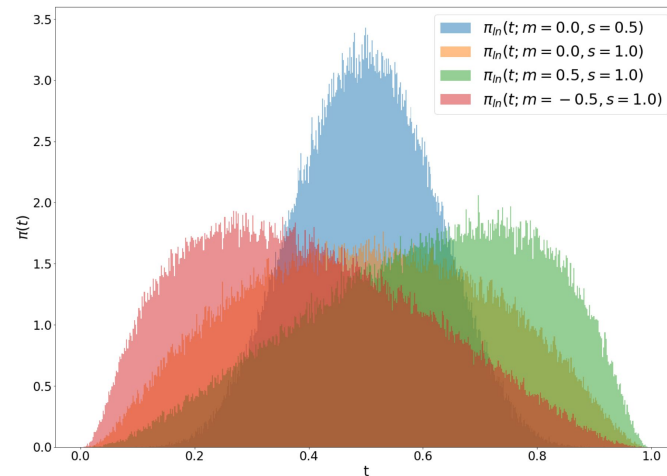
Goal. Emphasize on "hard tasks" at training time

Idea. Use different distribution for timesteps

Logit-Normal distribution

$$\pi_{\text{ln}}(t; m, s) = \frac{1}{s\sqrt{2\pi}t(1-t)} \exp\left(-\frac{(\text{logit}(t) - m)^2}{2s^2}\right)$$

where $\text{logit}(t) = \log\left(\frac{t}{1-t}\right)$ with $t \in (0, 1)$



Timestep adjustment with respect to resolution

Same noise level, different resolutions



Low resolution



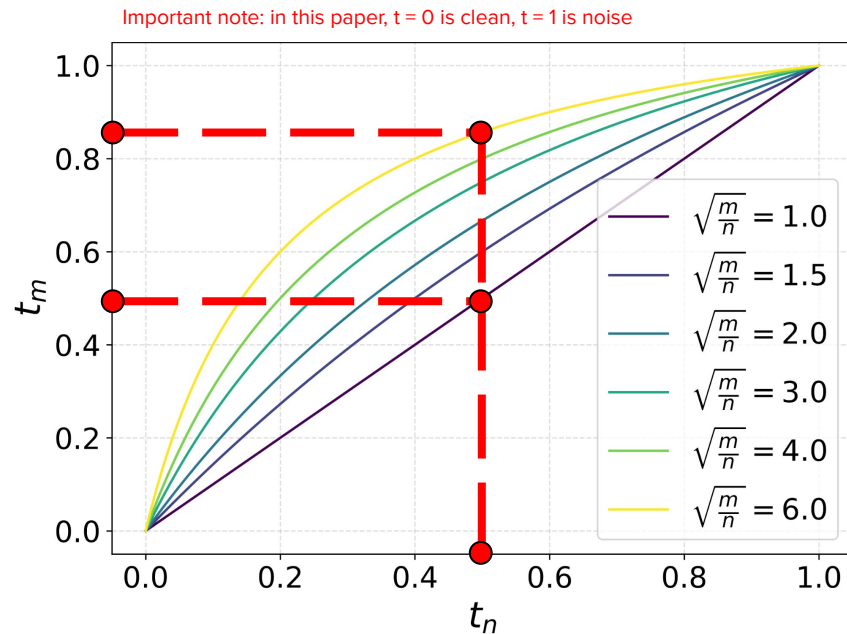
High resolution

Timestep adjustment with respect to resolution

At **higher** resolution:

- Same noise level leads to less "perceived noise"
- Solution: **shift timestep**

Shifted $t_m = \frac{\sqrt{\frac{m}{n}} t_n}{1 + \left(\sqrt{\frac{m}{n}} - 1\right) t_n}$ Original

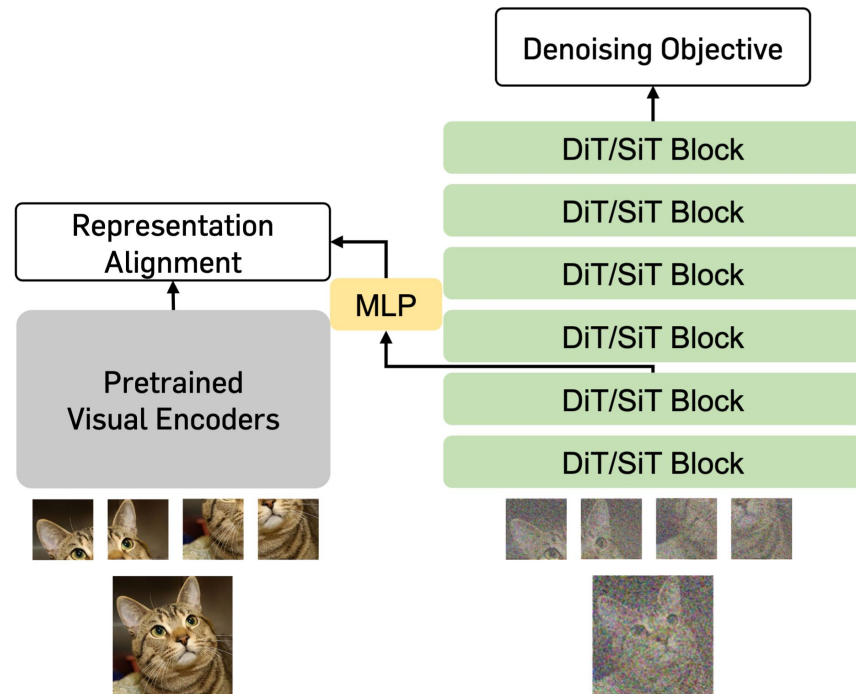


Representation alignment

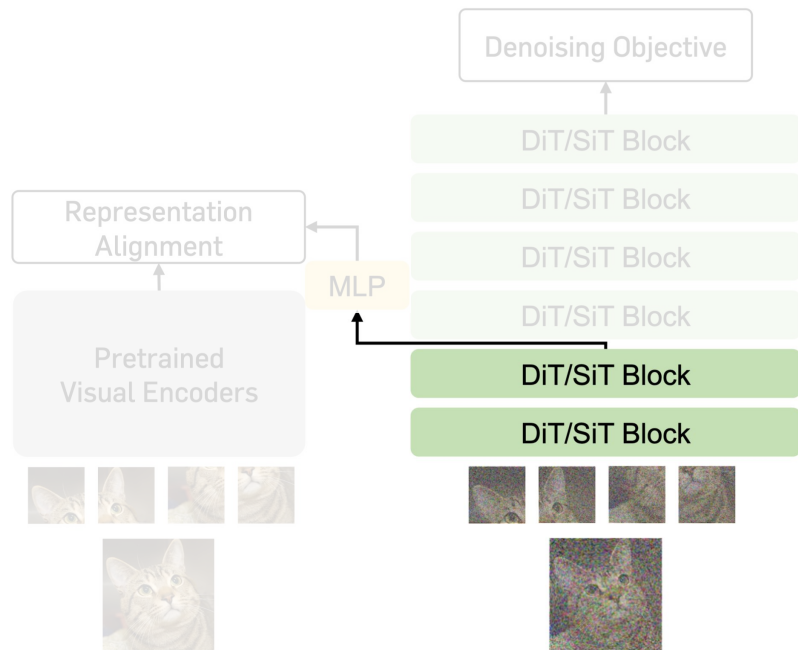
REPA = REPresentation Alignment

Idea. Leverage representation learned from big pretrained models

$$\mathcal{L} := \mathcal{L}_{\text{velocity}} + \lambda \mathcal{L}_{\text{REPA}}$$



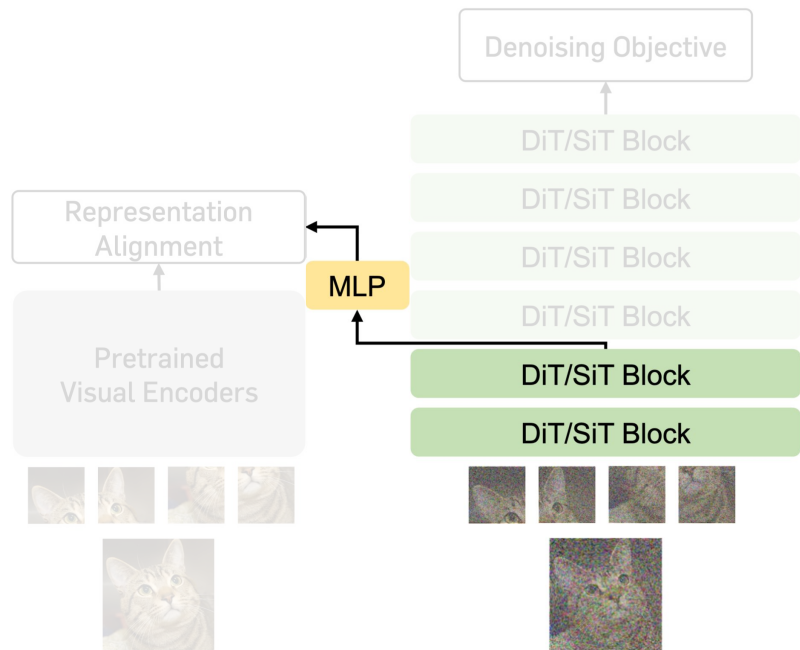
REPA loss



$\mathbf{h}_t^{[n]}$

Transformer
encoder output of
generation model

REPA loss

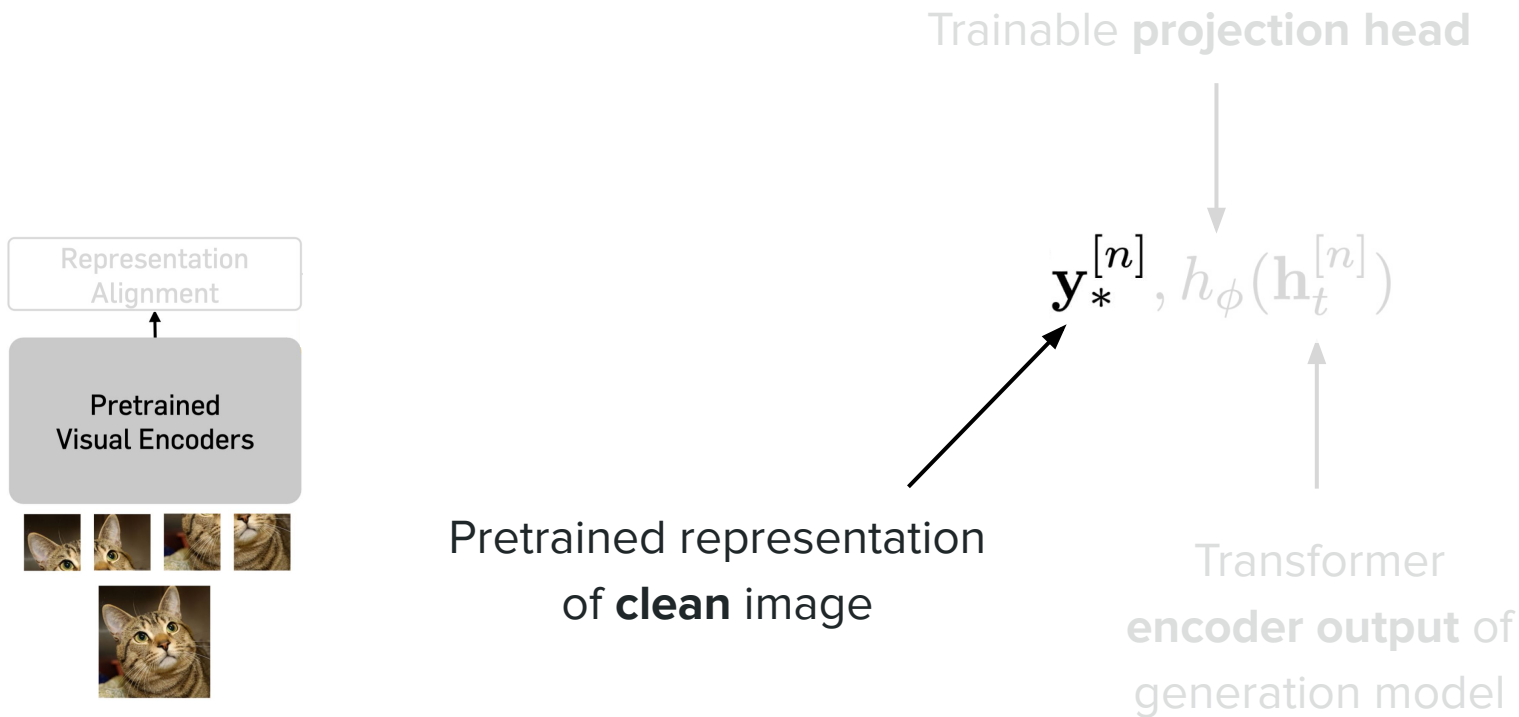


Trainable **projection head**

$$h_{\phi}(\mathbf{h}_t^{[n]})$$

Transformer
encoder output of
generation model

REPA loss



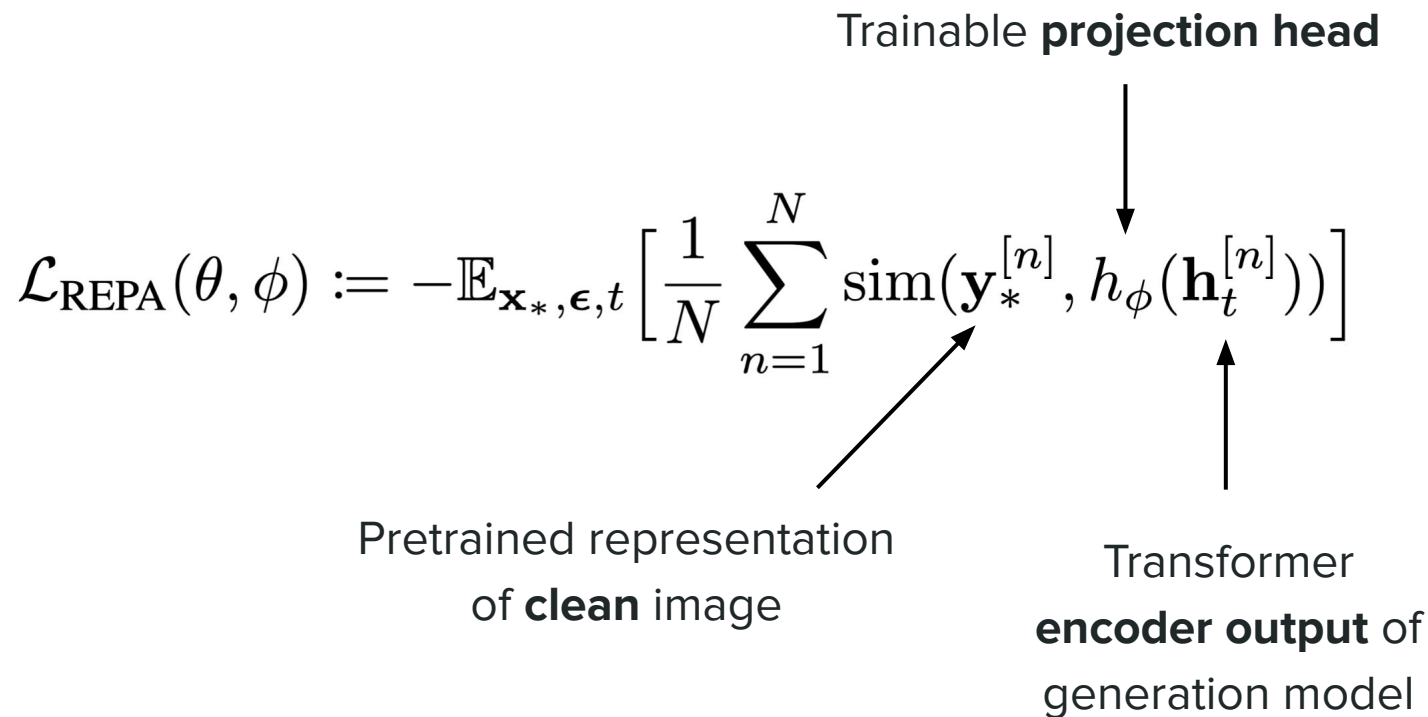
REPA loss

Trainable **projection head**

$$\mathcal{L}_{\text{REPA}}(\theta, \phi) := -\mathbb{E}_{\mathbf{x}_*, \epsilon, t} \left[\frac{1}{N} \sum_{n=1}^N \text{sim}(\mathbf{y}_*^{[n]}, h_{\phi}(\mathbf{h}_t^{[n]})) \right]$$

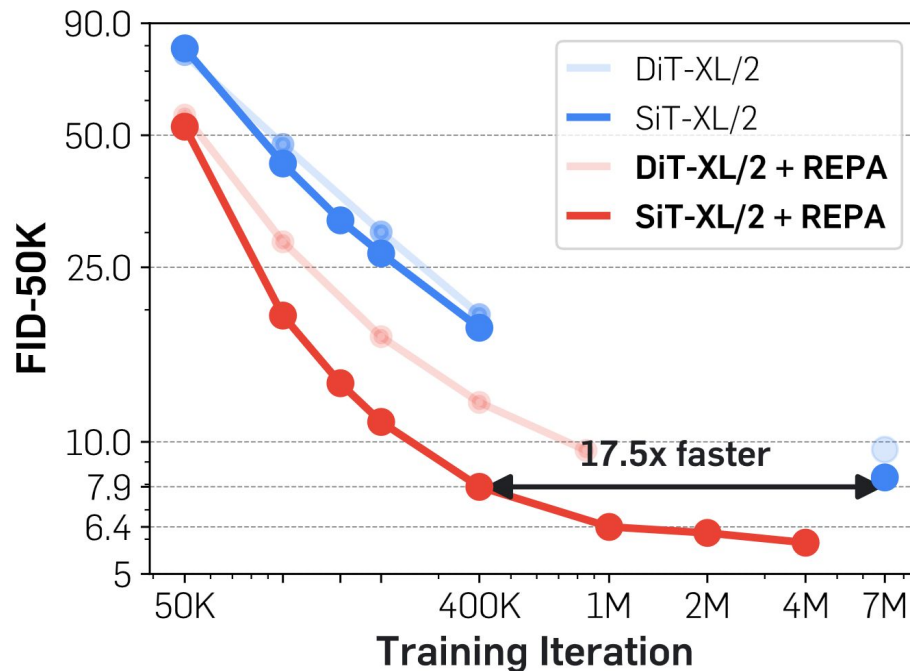
Pretrained representation
of **clean** image

Transformer
encoder output of
generation model



REPA results

- Significant **speedup**
- Only training with **first layers** leads to best performance
- Bigger diffusion models benefit even more from REPA





Diffusion & Large Vision Models

Lifecycle overview

Training loss

Pre-training

Post-training

Tuning

Distillation

Objective of pre-training

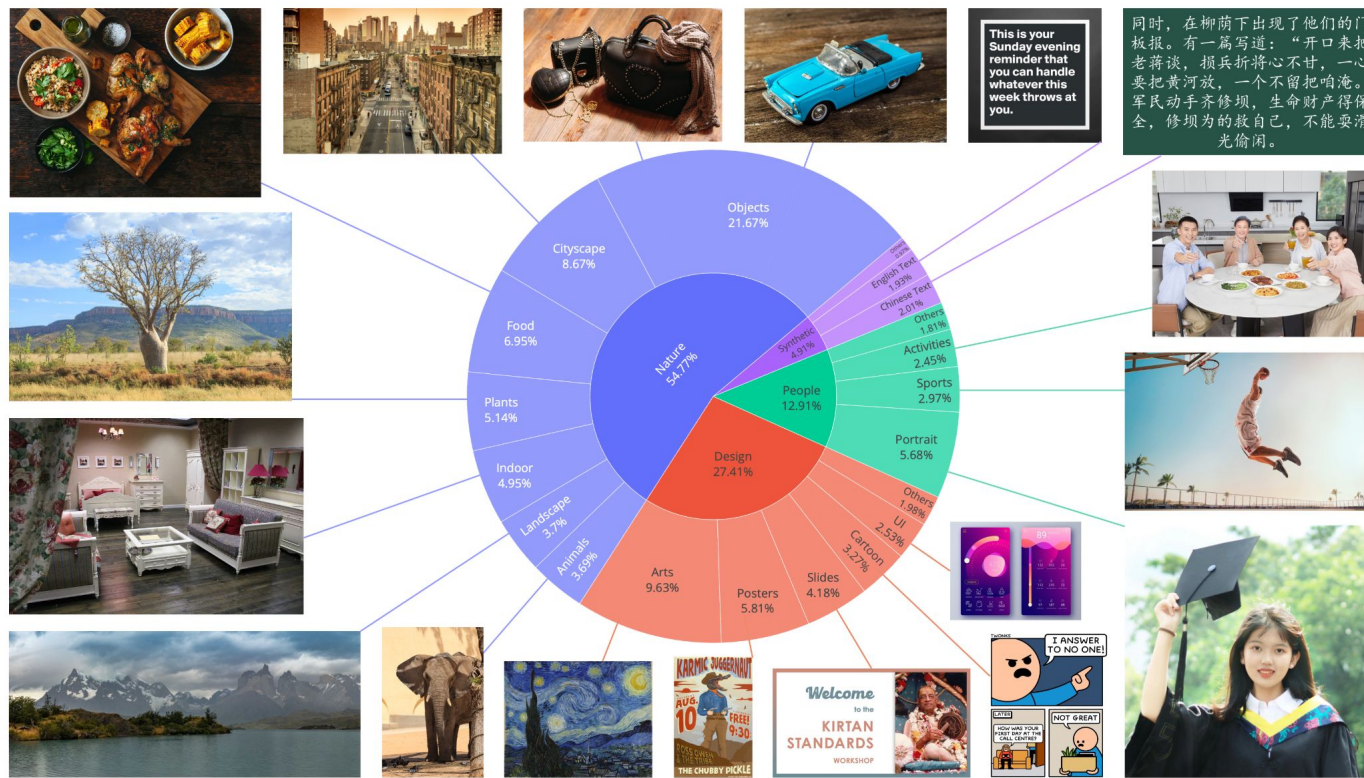
1

Pre-training



Learn **how** to
generate images

Data mixtures



Curriculum learning

EASY

- Low resolution
- Fixed squares
- Simple and short prompts

a teddy bear



Curriculum learning

EASY

- Low resolution
- Fixed squares
- Simple and short prompts

a teddy bear



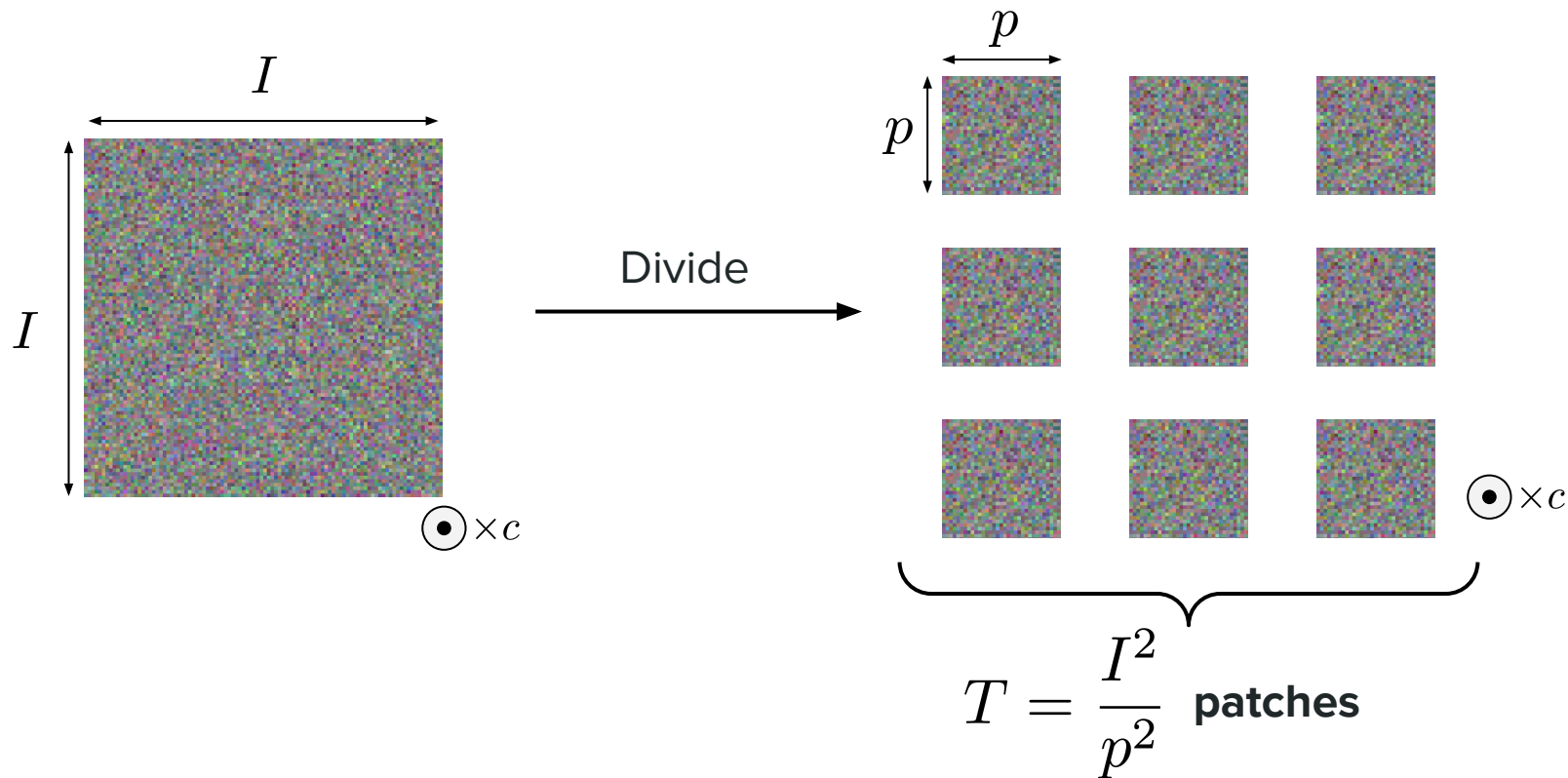
HARD

- High resolution
- Variable aspect ratios
- Complex, sophisticated prompts

A plush teddy bear in a beige trench coat, plaid scarf, and crossbody bag walks along a rain-soaked Parisian boulevard at dusk. Warm boutique lights, elegant Haussmann architecture, glowing streetlamps, and wet pavement reflections create a cinematic, cozy luxury travel scene.

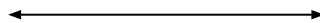
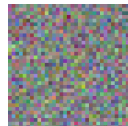


Handling different resolutions



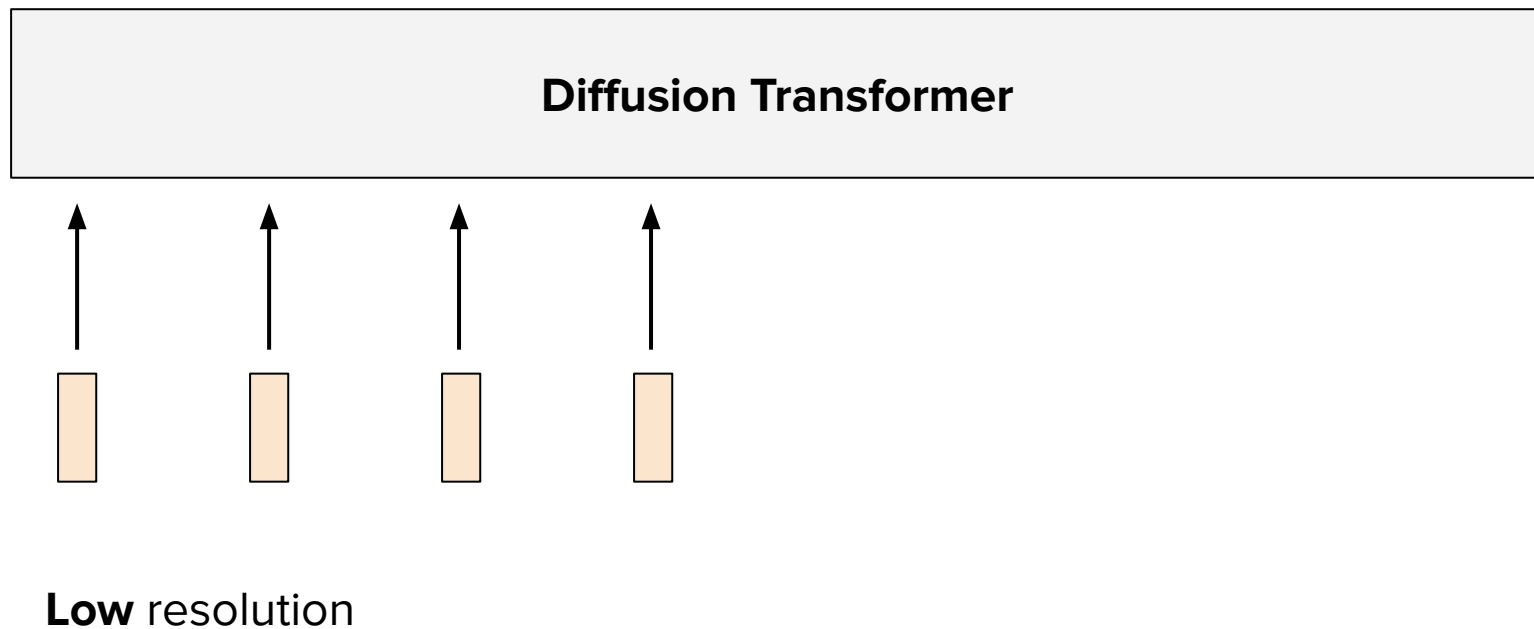
Handling different resolutions

Patch

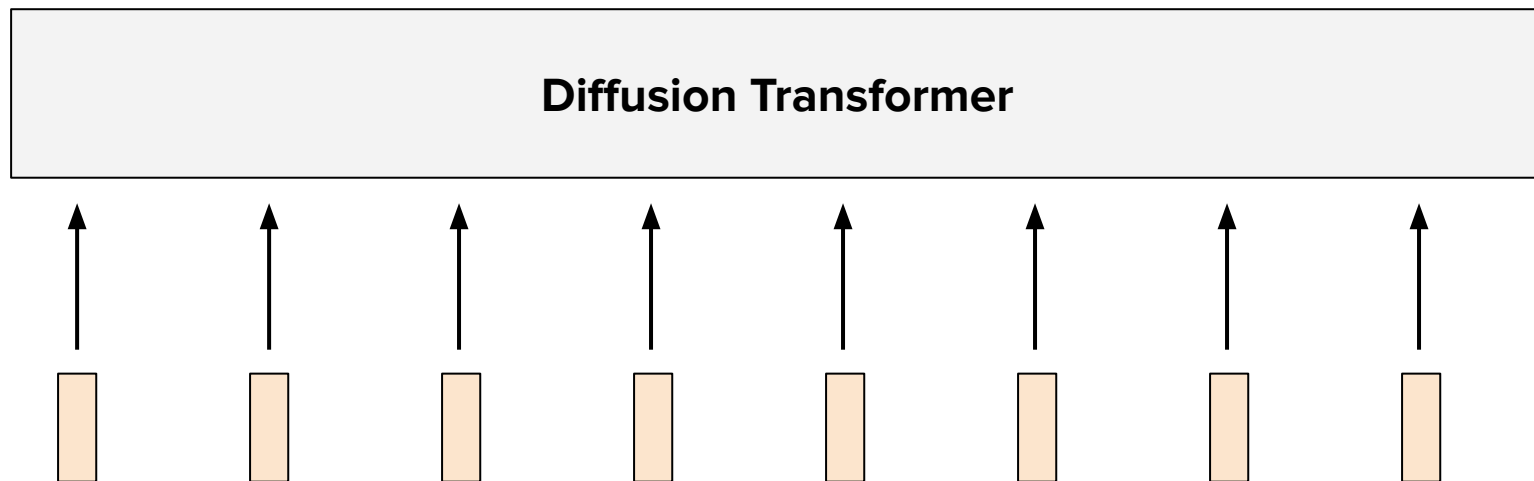


Embedding

Handling different resolutions



Handling different resolutions



High resolution



Diffusion & Large Vision Models

Lifecycle overview

Training loss

Pre-training

Post-training

Tuning

Distillation

Objective of post-training

1

Pre-training



Learn **how** to
generate images

2

Post-training



Learn how to
generate **good**
images

Continued training and supervised finetuning

Idea. Improve quality of generated images

**Continued Training
(CT)**

Focus on **knowledge**



Continued training and supervised finetuning

Idea. Improve quality of generated images

Continued Training (CT)

Focus on **knowledge**



Supervised FineTuning (SFT)

Focus on **behavior**



- Lighting
- Aesthetics
- Instruction following
among others

Preference tuning overview

Idea. Incorporate signal from human preference in way images are generated



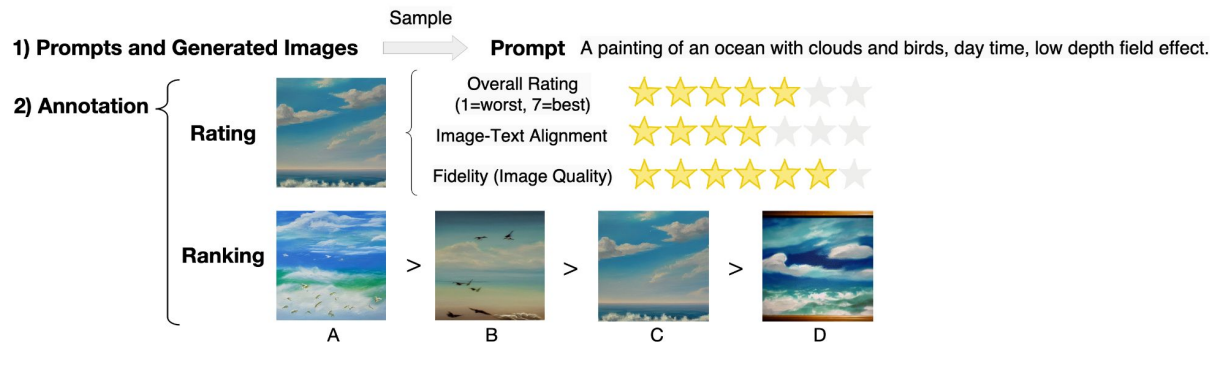
Preference tuning methods

- **Reward Feedback Learning (ReFL)**

1) **Prompts and Generated Images**  **Prompt** A painting of an ocean with clouds and birds, day time, low depth field effect.

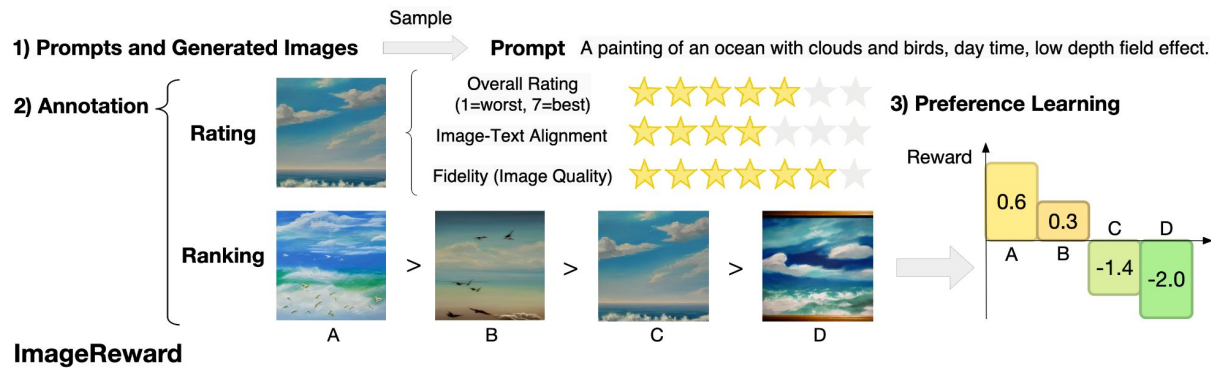
Preference tuning methods

- **Reward Feedback Learning (ReFL)**



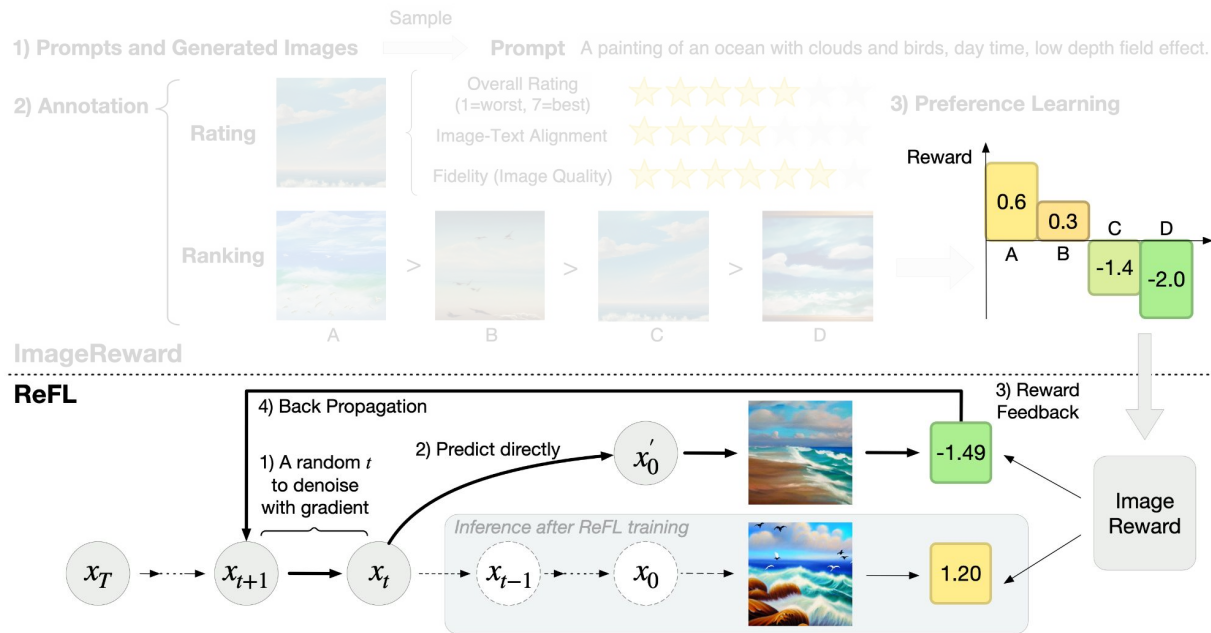
Preference tuning methods

- **Reward Feedback Learning (ReFL)**



Preference tuning methods

- **Reward Feedback Learning (ReFL)**



Preference tuning methods

- Reward Feedback Learning (ReFL)
- **Flow-Group Reward Policy Optimization (Flow-GRPO)**

Prompt: A photo of four cups.

Preference tuning methods

- Reward Feedback Learning (ReFL)
- **Flow-Group Reward Policy Optimization (Flow-GRPO)**

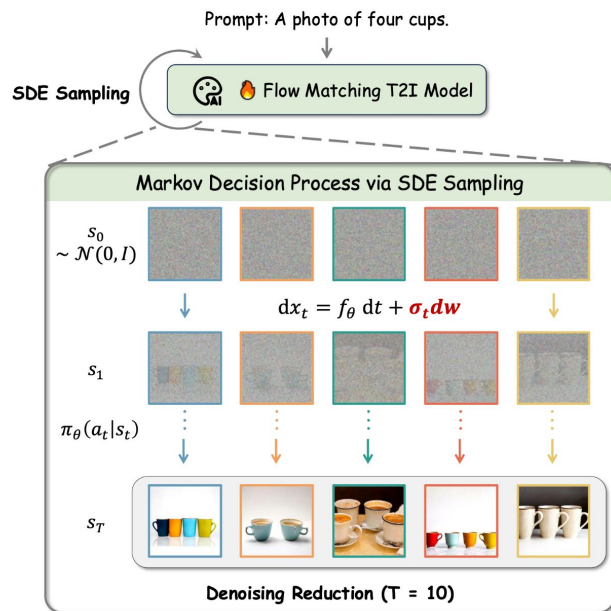
Prompt: A photo of four cups.



Flow Matching T2I Model

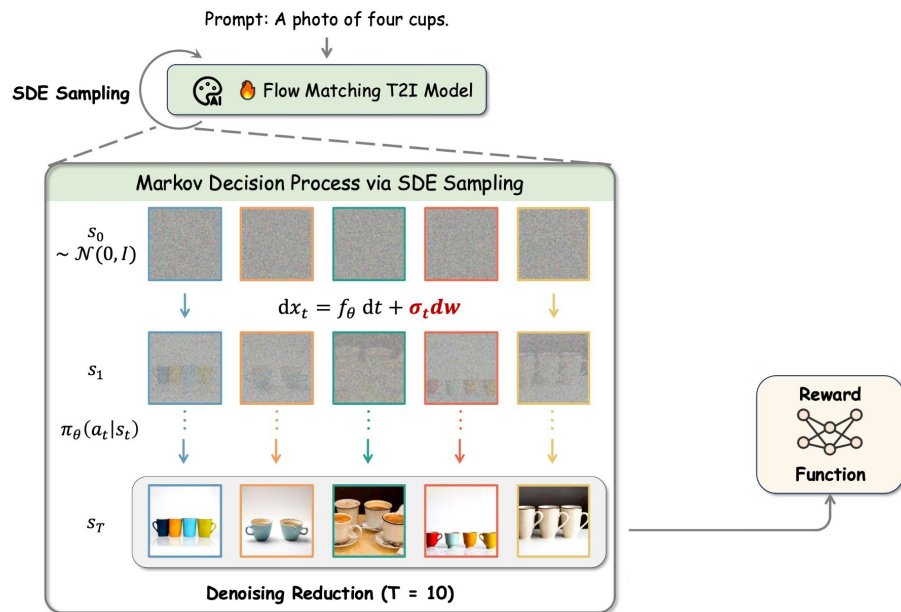
Preference tuning methods

- Reward Feedback Learning (ReFL)
- **Flow-Group Reward Policy Optimization (Flow-GRPO)**



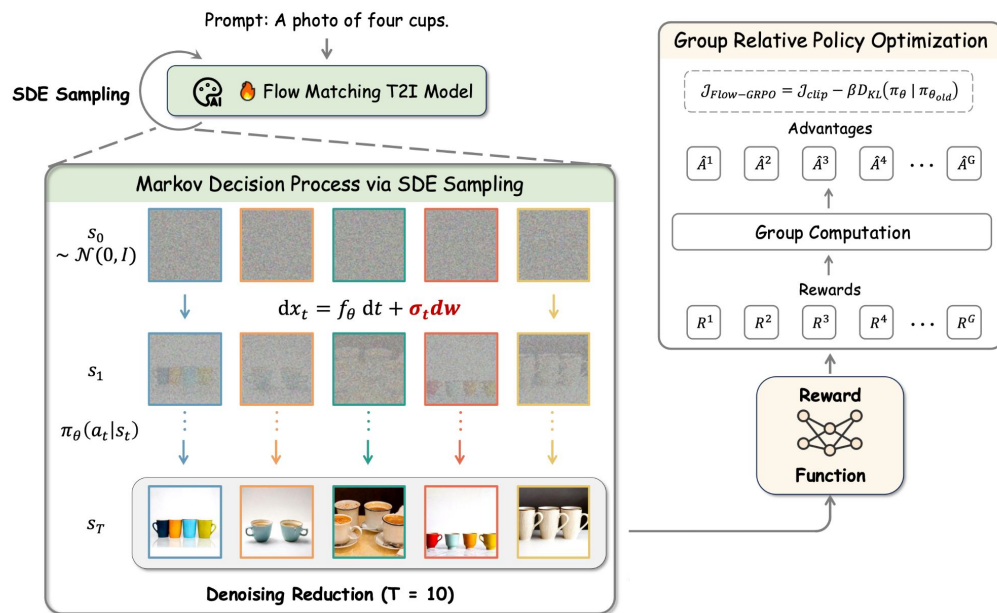
Preference tuning methods

- Reward Feedback Learning (ReFL)
- **Flow-Group Reward Policy Optimization (Flow-GRPO)**



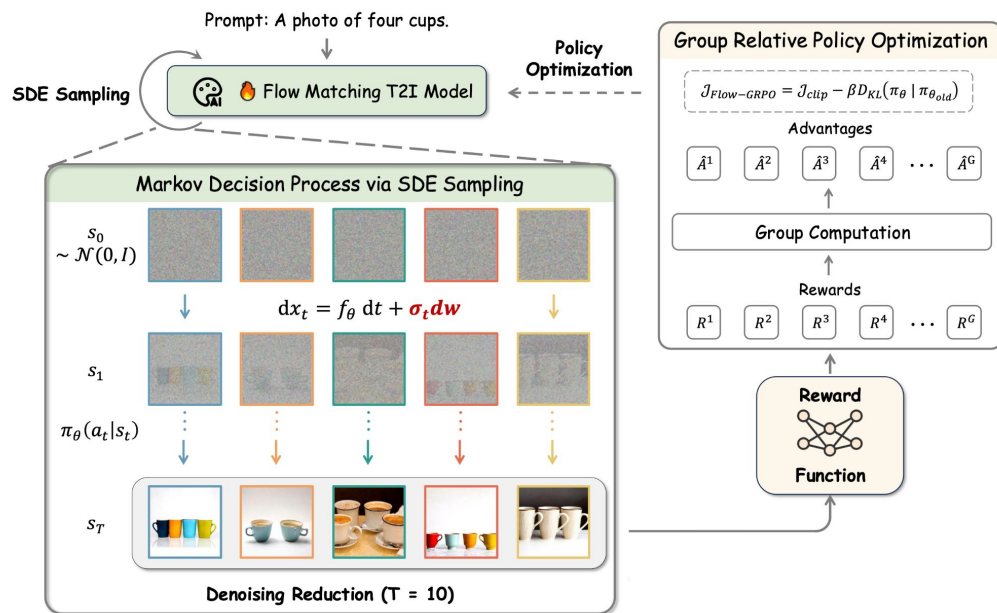
Preference tuning methods

- Reward Feedback Learning (ReFL)
- **Flow-Group Reward Policy Optimization (Flow-GRPO)**



Preference tuning methods

- Reward Feedback Learning (ReFL)
- **Flow-Group Reward Policy Optimization (Flow-GRPO)**



Preference tuning methods

- Reward Feedback Learning (ReFL)
- Flow-Group Reward Policy Optimization (Flow-GRPO)
- **Diffusion-Direct Preference Optimization (Diffusion-DPO)**

$$\text{Diff}_{\text{policy}} = \left(\|v_{\theta}(x_t^{\text{win}}, h, t) - v_t^{\text{win}}\|_2^2 - \|v_{\theta}(x_t^{\text{lose}}, h, t) - v_t^{\text{lose}}\|_2^2 \right)$$

$$\text{Diff}_{\text{ref}} = \left(\|v_{\text{ref}}(x_t^{\text{win}}, h, t) - v_t^{\text{win}}\|_2^2 - \|v_{\text{ref}}(x_t^{\text{lose}}, h, t) - v_t^{\text{lose}}\|_2^2 \right)$$

$$\mathcal{L}_{DPO} = -\mathbb{E}_{h, (x_0^{\text{win}}, x_0^{\text{lose}}) \sim \mathcal{D}, t \sim \mathcal{U}(0,1)} \left[\log \sigma \left(-\beta (\text{Diff}_{\text{policy}} - \text{Diff}_{\text{ref}}) \right) \right]$$

Optimizing input prompt



Optimizing input prompt

PE = Prompt Enhancement



A warm, intimate indoor reading scene featuring a plush teddy bear as the central subject. The teddy bear is seated comfortably in a soft, textured armchair, positioned slightly angled toward the viewer. It has light beige fur with realistic, finely detailed fibers, giving it a soft and touchable appearance. The bear wears a cozy, knitted brown sweater with visible wool texture and subtle stitching imperfections for realism. Perched on its face are small, round, thin-wire eyeglasses, slightly oversized, resting naturally on its snout. The teddy bear is holding an open hardcover book titled "The Tales of Woodland", with an elegant, muted green cover and gold embossed serif typography. The book appears slightly worn but well cared for, with visible page thickness and gentle curvature where it is being held. The bear's gaze is directed downward toward the book, creating a calm, focused, and contemplative mood. Lighting is soft, warm, and cinematic, coming primarily from a table lamp positioned to the left of the frame. The lamp emits a golden, diffused glow, casting gentle shadows and creating a cozy, evening ambiance. The lighting should produce a shallow depth-of-field effect, with the subject sharply in focus while the background softly blurs (bokeh effect). In the foreground, a ceramic mug sits on a wooden side table. The mug is off-white with a subtle illustration of a bear printed on it, reinforcing the theme. The mug contains a warm beverage, with faint steam rising, suggesting warmth and comfort. The wooden table has a natural grain texture, slightly worn edges, and a matte finish...[truncated]



Diffusion & Large Vision Models

Lifecycle overview

Training loss

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Tuning

Distillation

Objective of tuning

1

Pre-training



Learn **how** to
generate images

2

Post-training



Learn how to
generate **good**
images

3

Tuning



Learn how to
generate images
for a special case

Using a few images for tuning...

DreamBooth

Training. Tune weights to associated subject with rare token

$$\mathcal{L} = \mathcal{L}_{\text{instance}} + \lambda \mathcal{L}_{\text{prior}}$$



a photo of a [V] teddy bear



a photo of a teddy bear

Using a few images for tuning...

DreamBooth

Training. Tune weights to associated subject with rare token

$$\mathcal{L} = \mathcal{L}_{\text{instance}} + \lambda \mathcal{L}_{\text{prior}} \leftarrow \text{Prevent overfitting}$$



a photo of a [V] teddy bear

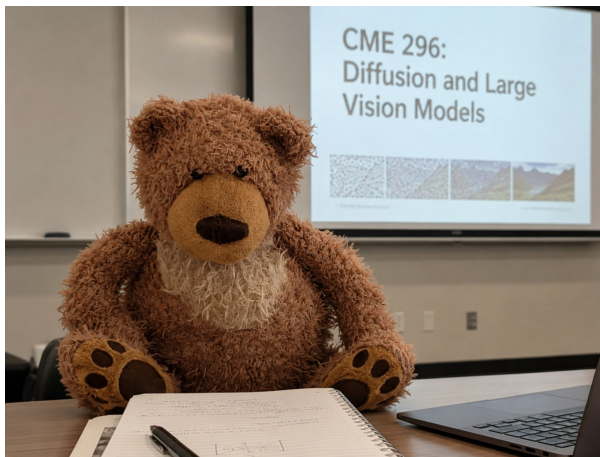


a photo of a teddy bear

Using a few images for tuning...

DreamBooth

Inference. Use rare token to generate image with subject



[V] teddy bear attending CME 296

...and only changing a small number of weights!

LoRA = Low Rank Adaptation

Context. Tuning is resource intensive and not everyone has big GPUs.

Idea. Low-Rank Adaptation (LoRA) approximates matrix with product of two low-rank matrices

$$W = W_0 + A \times B$$

The diagram shows the LoRA equation: $W = W_0 + A \times B$. W is represented by a blue rectangle. W_0 is represented by a light gray rectangle. A is represented by a tall green rectangle, and B is represented by a wide green rectangle. The equation is shown with an equals sign and a plus sign, and the multiplication is indicated by a cross symbol.

Discussion. Fraction of parameters need to be trained with similar performance

Discussion

Pros

-  Minimal overhead for inference
-  High fidelity compared to ref images

Cons

-  Training is time-consuming and costly
-  Not really reusable

Discussion

Pros

-  Minimal overhead for inference
-  High fidelity compared to ref images

Cons

-  Training is time-consuming and costly
-  Not really reusable

Use case. Great method if we want to **generate many** images



Diffusion & Large Vision Models

Lifecycle overview

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Objective of distillation

1

Pre-training



Learn **how** to
generate images

2

Post-training



Learn how to
generate **good**
images

3

Tuning



Learn how to
generate images
for a special case

4

Distillation



Learn how to
generate images,
fast

Problem statement

Motivation. Real-life use cases:

- Interactions
- High-volume
- Animations

Problem statement

Motivation. Real-life use cases:

- Interactions
- High-volume
- Animations

Constraints. Everyday users typically have:

- Limited compute
- Limited money
- Limited time

Distillation

Mindset.

- Trade-off quality / speed
- Leverage rich outputs from bigger, slower models
- Teacher vs. student

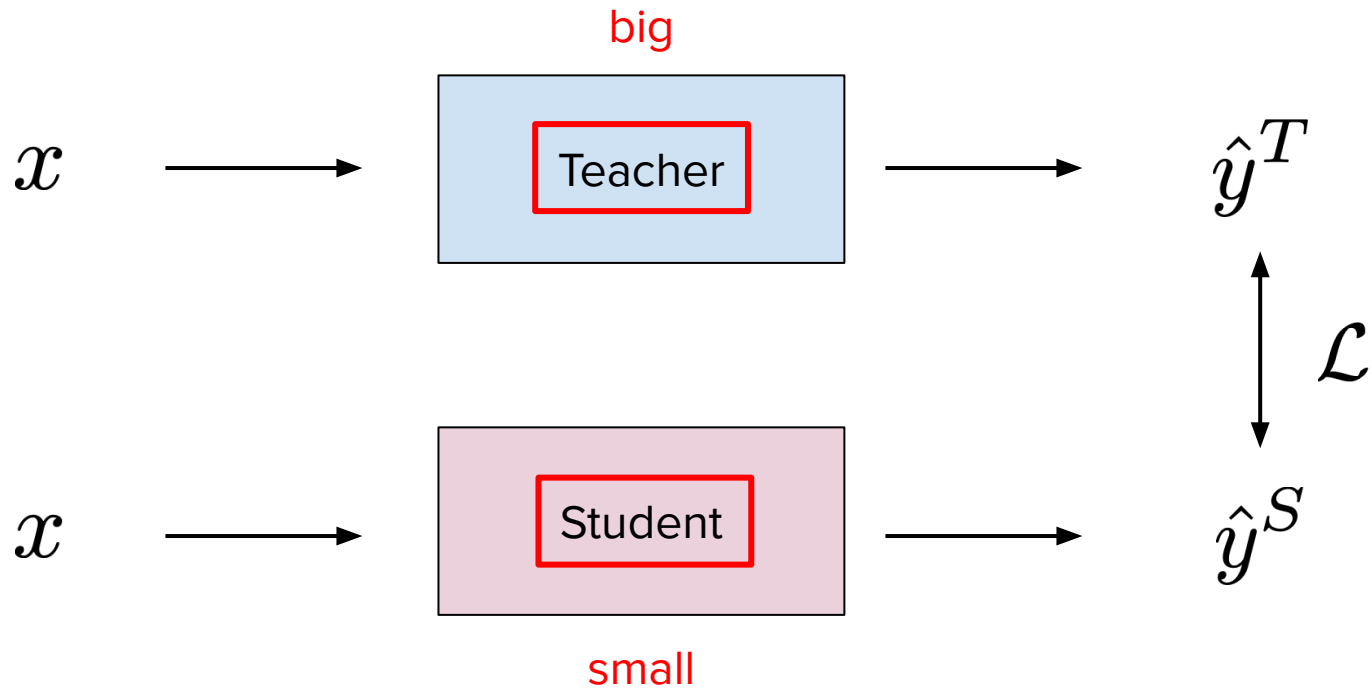
Distillation

Mindset.

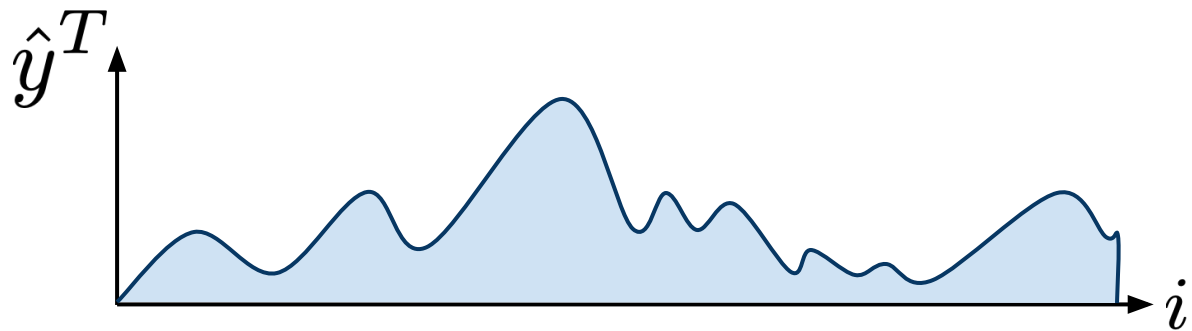
- Trade-off quality / speed
- Leverage rich outputs from bigger, slower models
- Teacher vs. student

Goal. Faster, cheaper inference.

“Traditional” distillation



“Traditional” distillation



$$\updownarrow \quad \text{KL}(\hat{y}^T \parallel \hat{y}^S) = \sum_i \hat{y}_{(i)}^T \log \left(\frac{\hat{y}_{(i)}^T}{\hat{y}_{(i)}^S} \right)$$

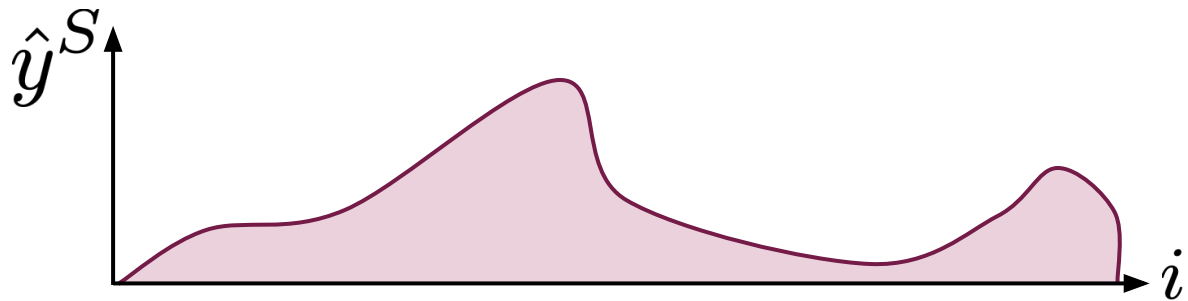


Image generation setup

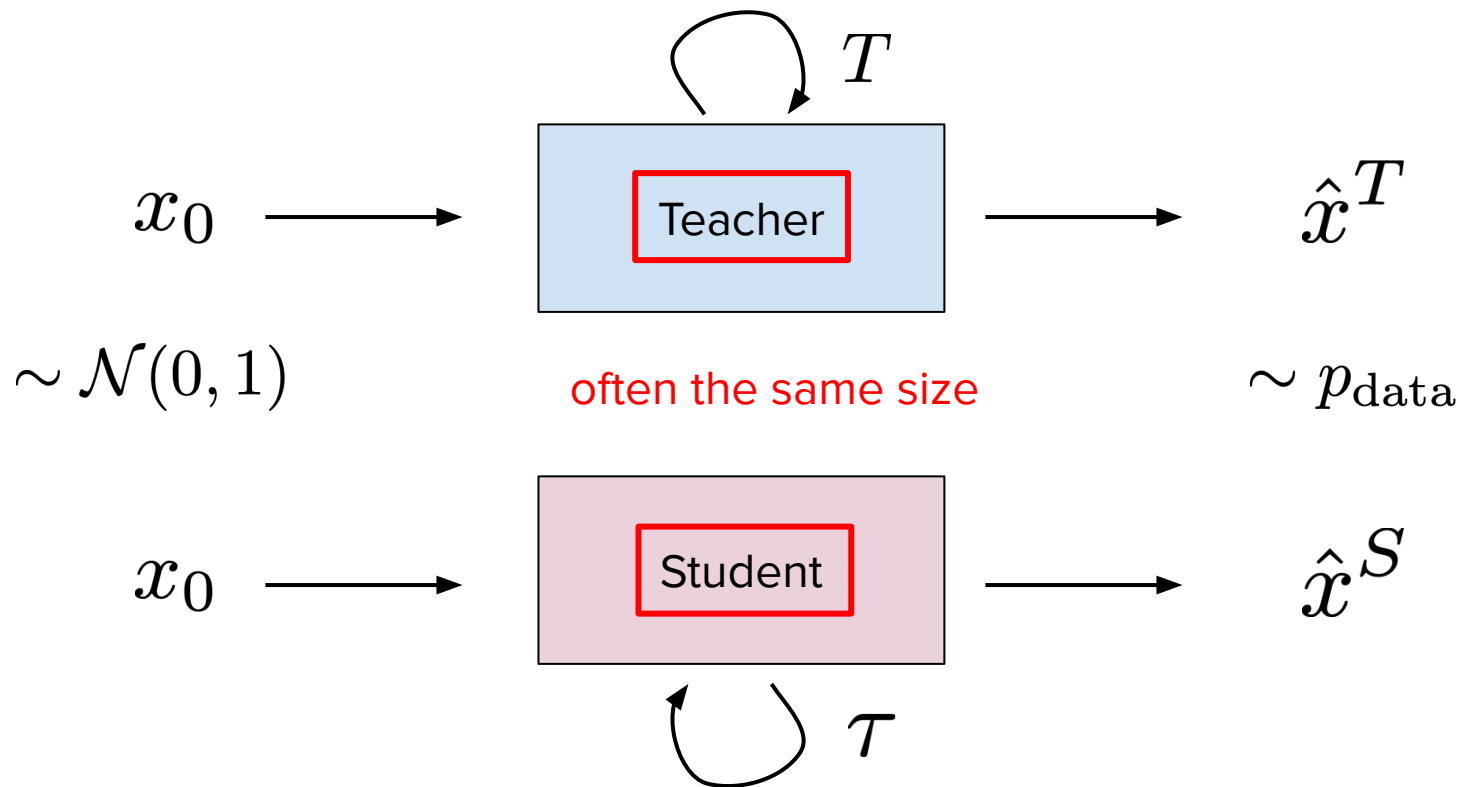


Image generation setup

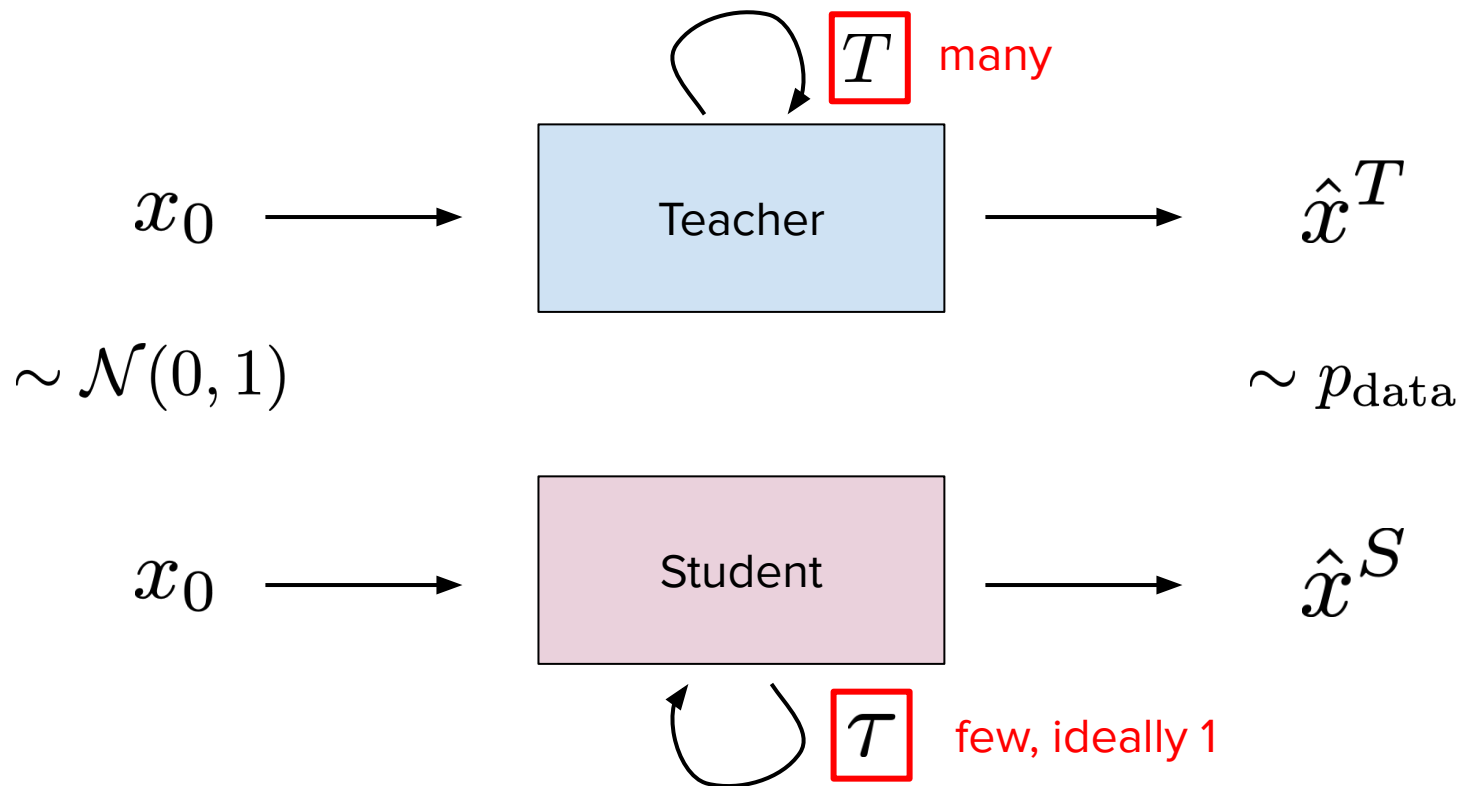
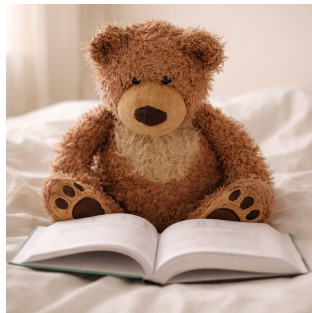


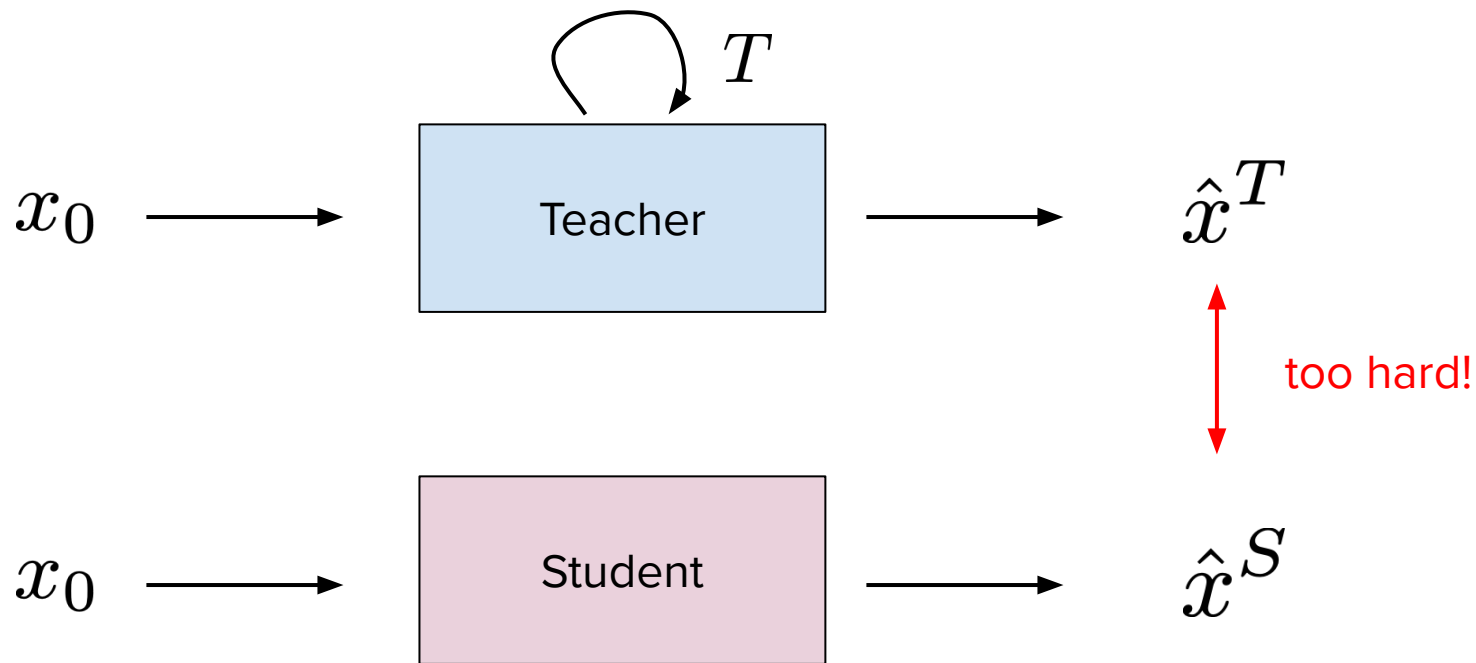
Image generation setup

 $\hat{x}^T$ 

$$||\hat{x}^T - \hat{x}^S||^2$$

 $\hat{x}^S$ 

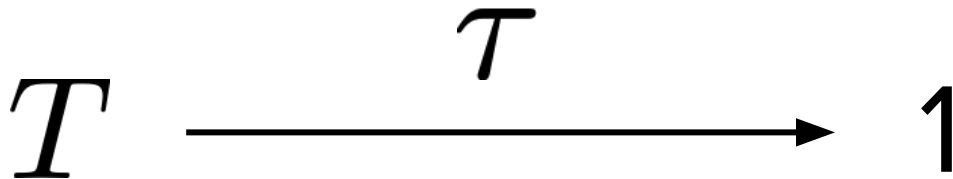
Initial attempt



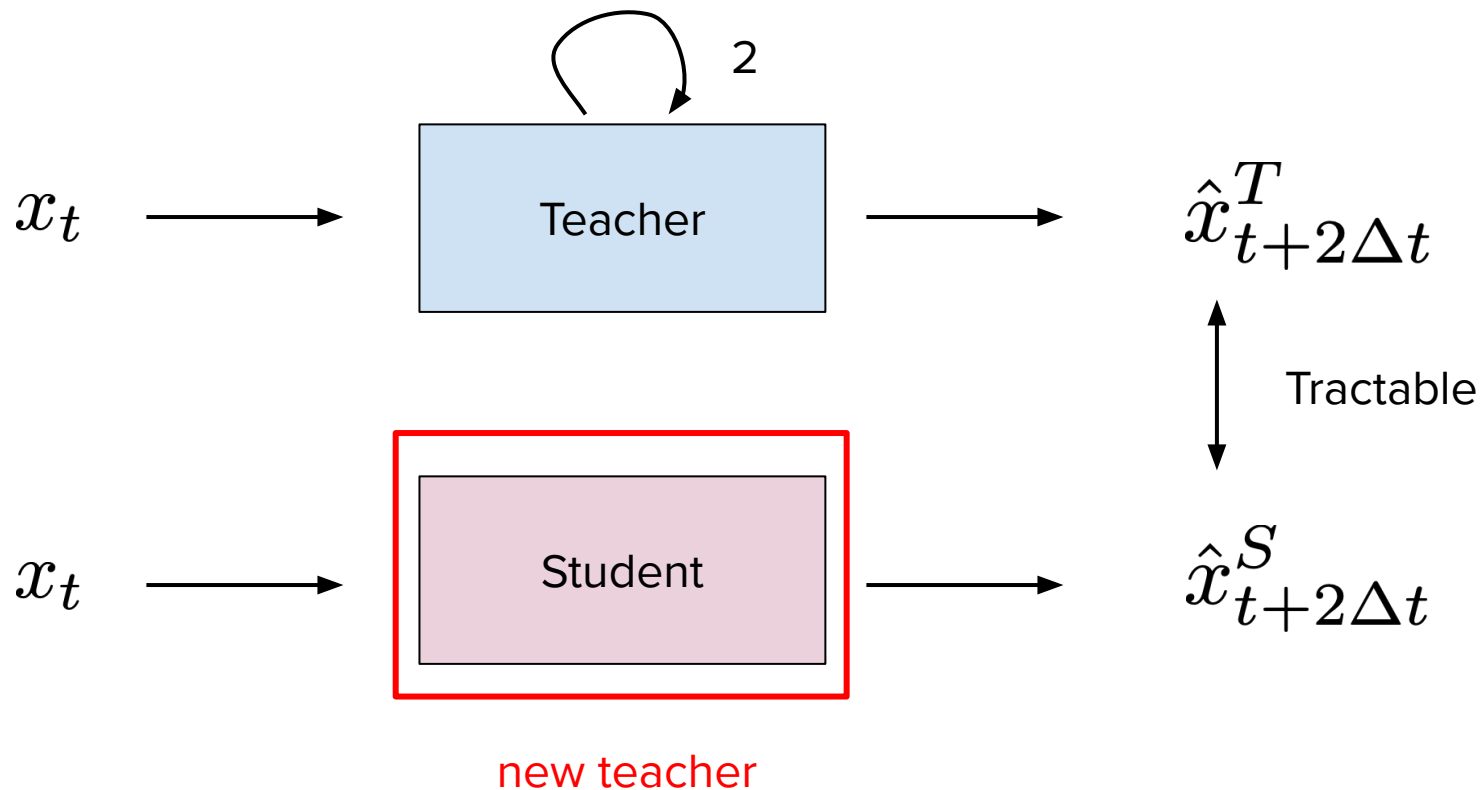
Progressive distillation

PD = **P**rogressive **D**istillation

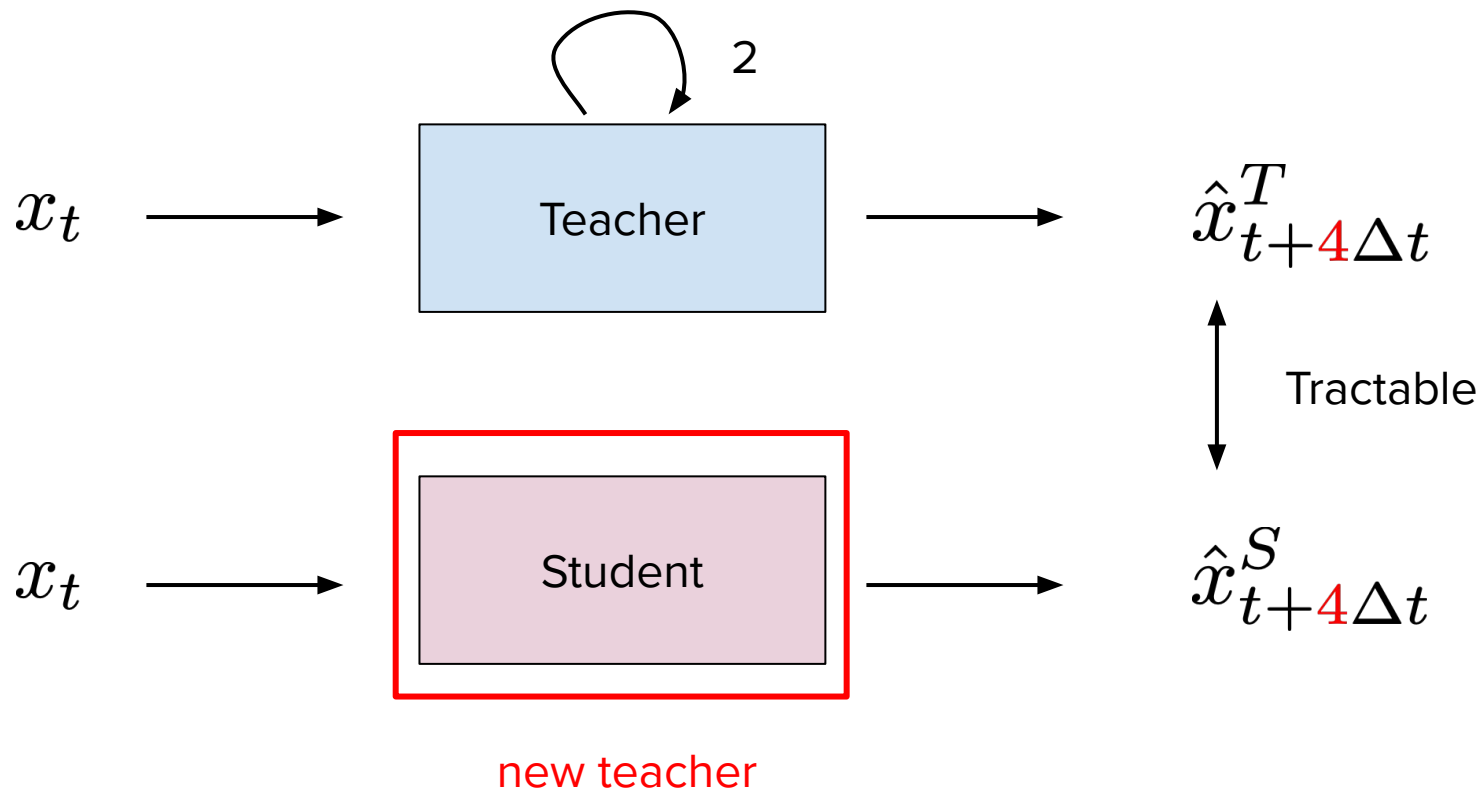
Remediation. Ask a teacher model to take several small steps and student to directly fit the transition.



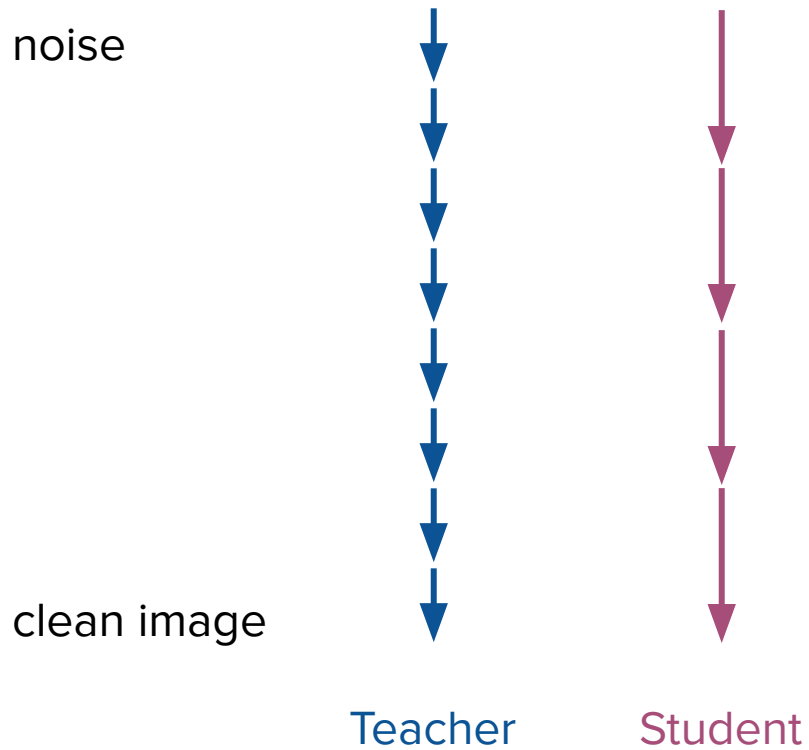
Progressive distillation



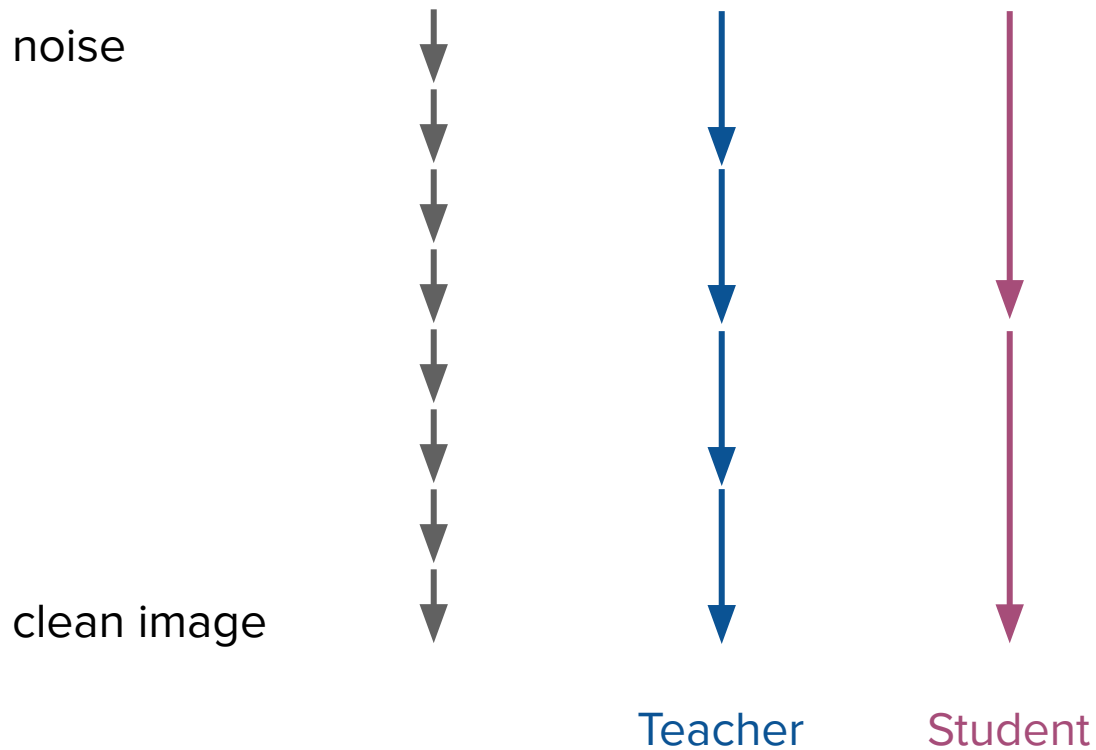
Progressive distillation



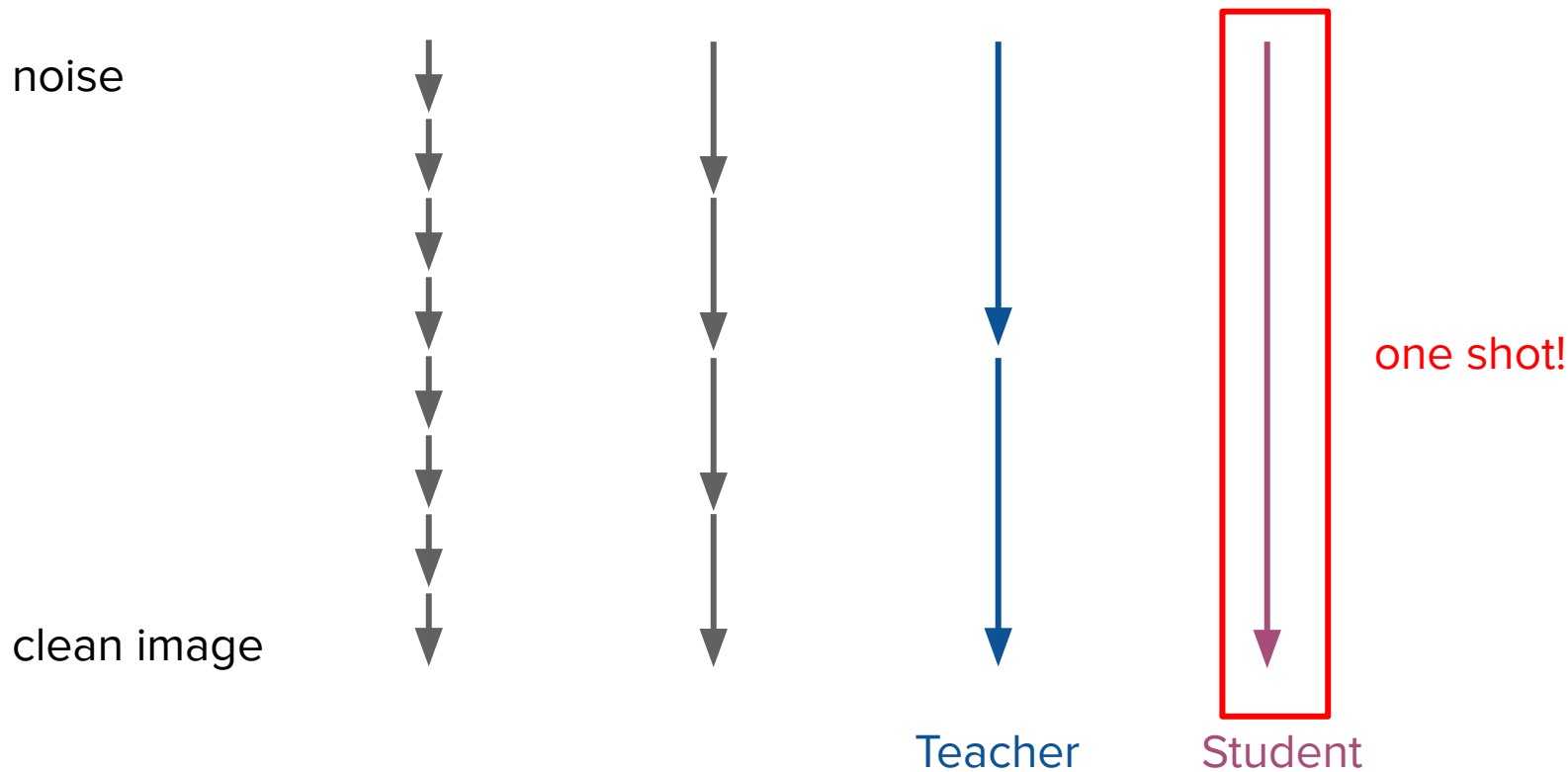
Progressive distillation



Progressive distillation

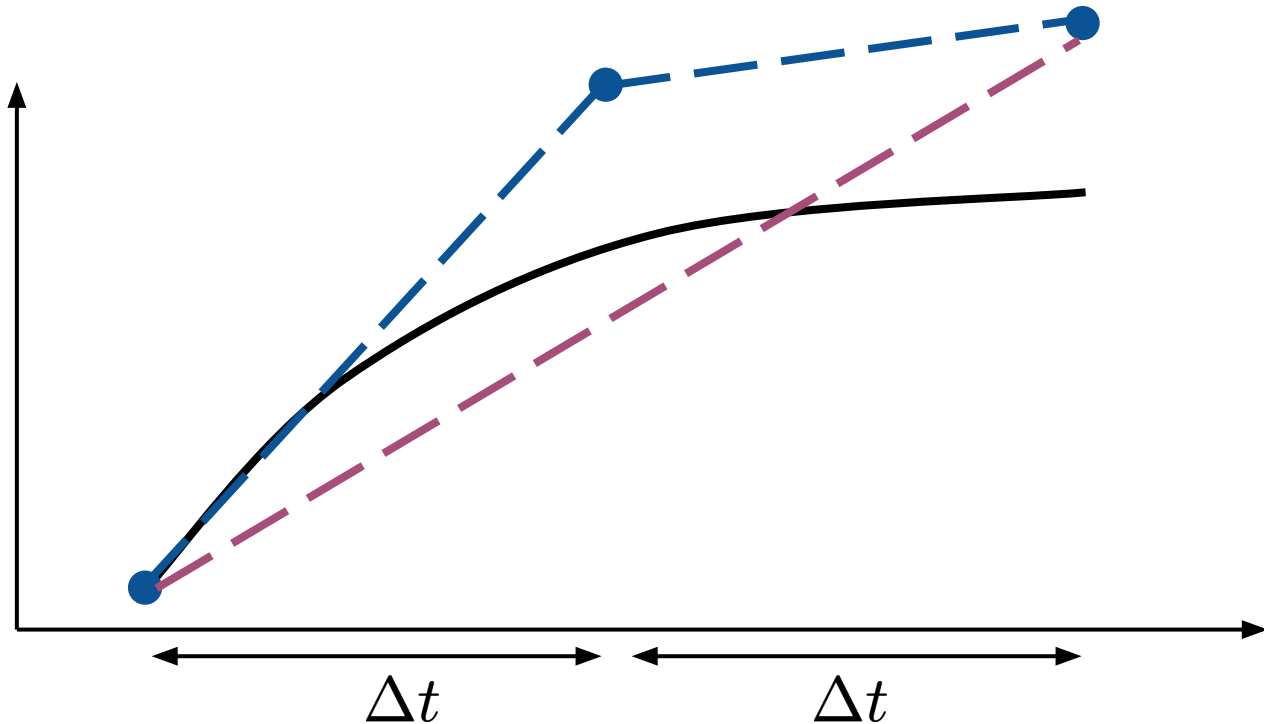


Progressive distillation



Progressive distillation

What is learned.

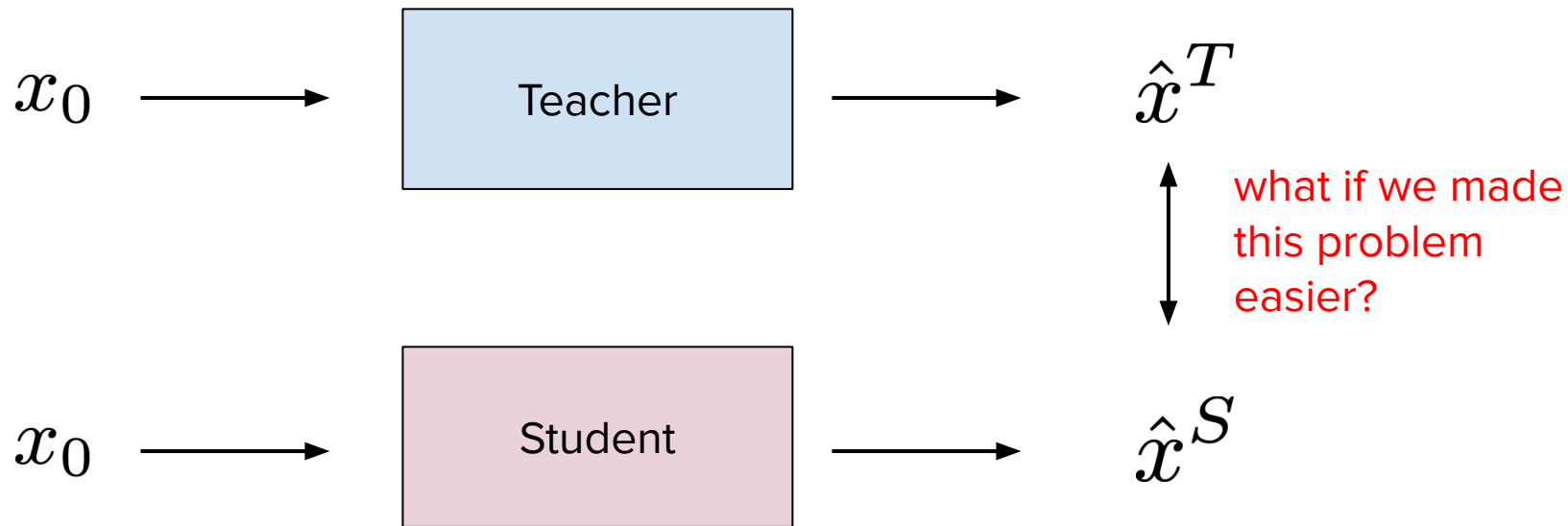


Progressive distillation

Results. Better quality than DDIM at iso-NFEs.

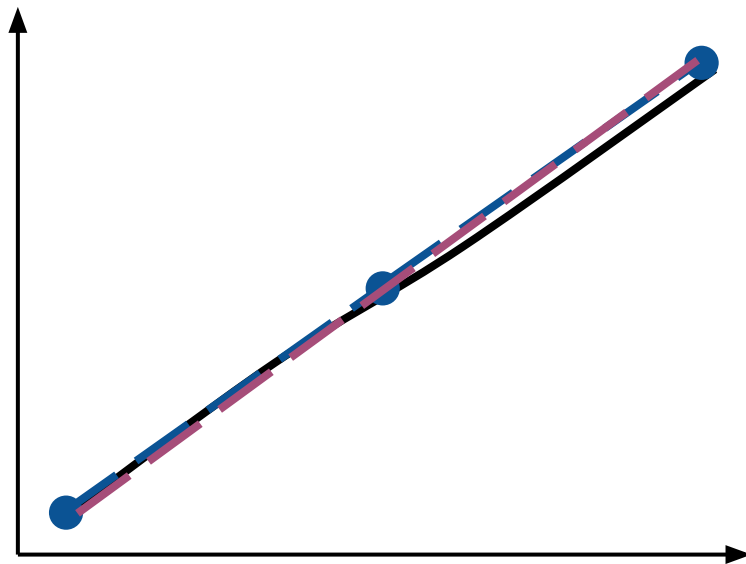
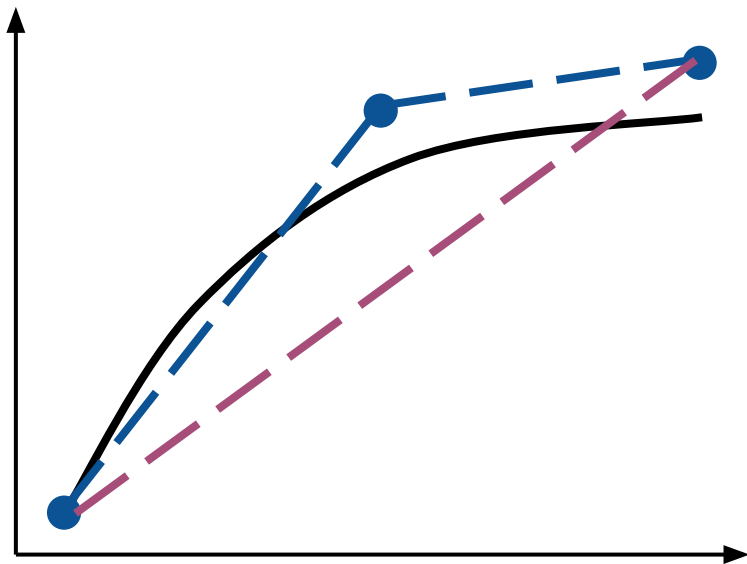
Method	Model evaluations	FID
Progressive Distillation (ours)	1	9.12
	2	4.51
	4	3.00
	8	2.57
Knowledge distillation (Luhman & Luhman, 2021)	1	9.36
DDIM (Song et al., 2021a)	10	13.36
	20	6.84
	50	4.67
	100	4.16

Progressive distillation



Progressive distillation

Wish.



Progressive distillation

Limitations.

- Curved paths = fixed constraints
- Requires discrete, fixed sampling at inference time
- Needs $\log(T)$ rounds of distillation

Progressive distillation

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- Curved paths = fixed constraints
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Attractive alternative mindset.

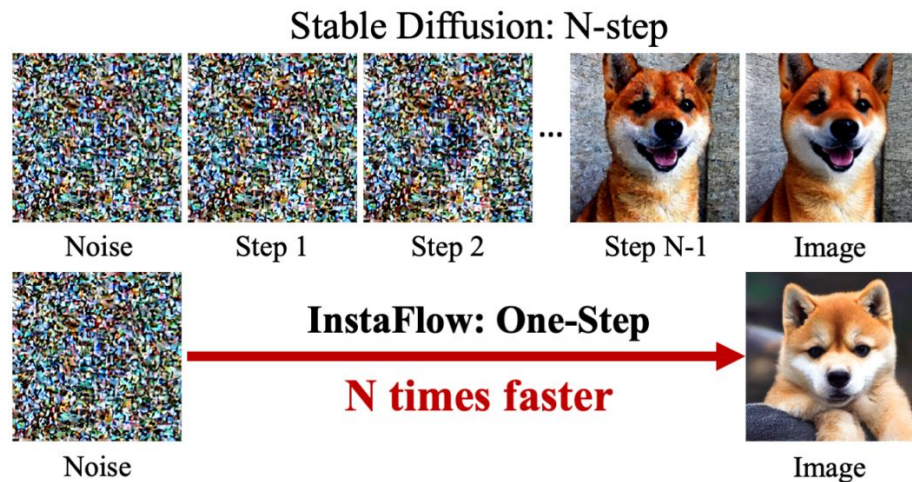
Original constraints  **Simplified** constraints

InstaFlow

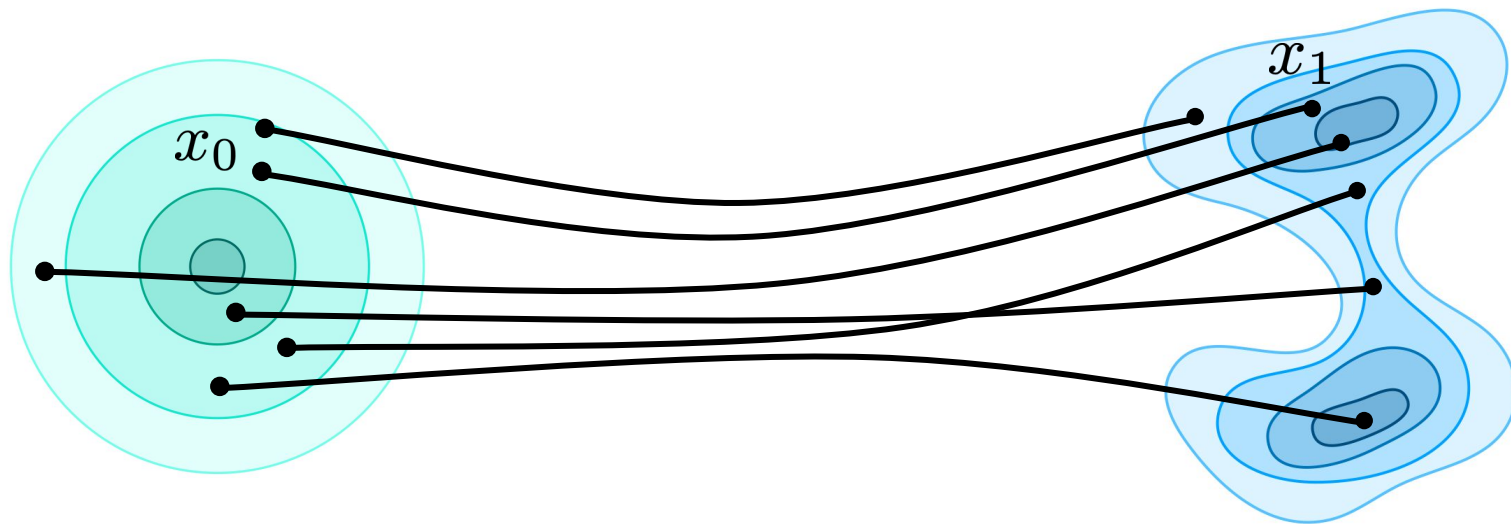
Idea. Leverage the "reflow" procedure

Process.

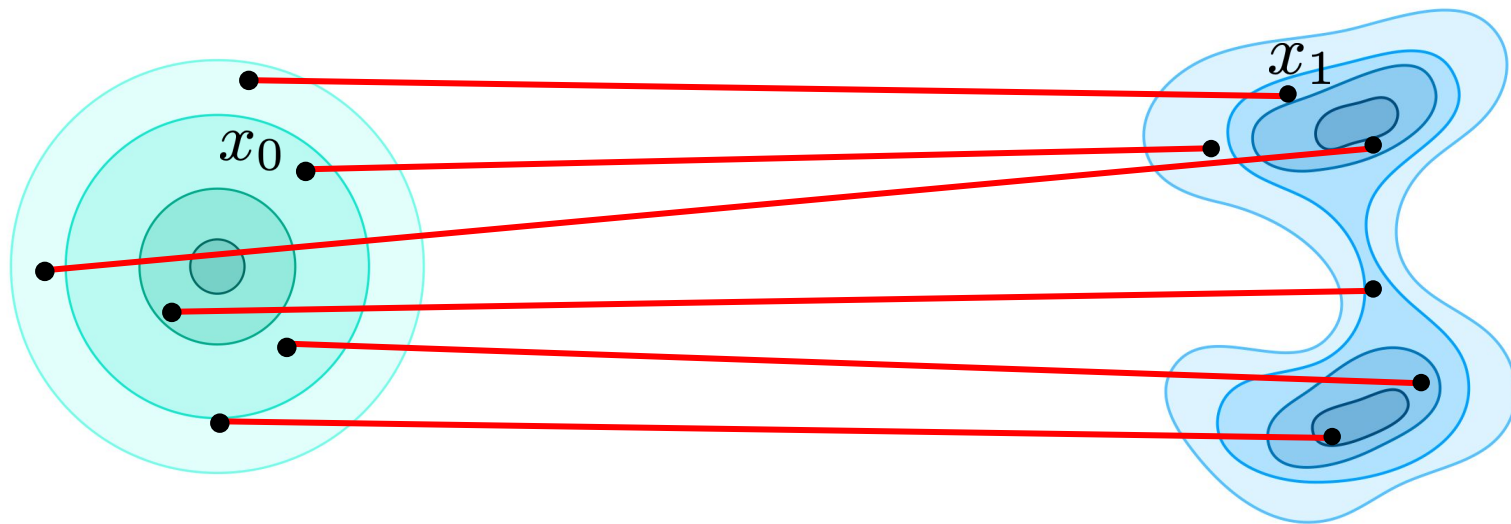
- Start from pretrained model
- Apply "reflow"
- Distill N-step model into a single-step one



Reflow. Before



Reflow. After



Distillation loss.

$$\tilde{v}_k = \arg \min_v \mathbb{E}_{X_0 \sim \pi_0, \mathcal{T} \sim D_{\mathcal{T}}} [\mathbb{D}(\text{ODE}[v_k](X_0 \mid \mathcal{T}), X_0 + v(X_0 \mid \mathcal{T}))]$$

teacher student

Distillation loss.

$$\tilde{v}_k = \arg \min_v \mathbb{E}_{X_0 \sim \pi_0, \mathcal{T} \sim D_{\mathcal{T}}} [\mathbb{D}(\text{ODE}[v_k](X_0 \mid \mathcal{T}), X_0 + v(X_0 \mid \mathcal{T}))]$$

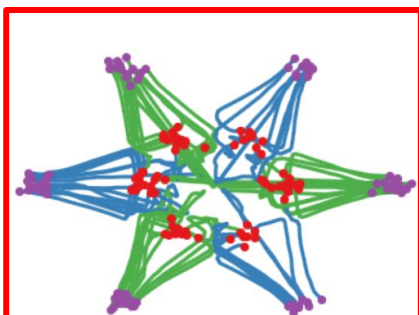
Distance function.

$$\mathbb{D}_{\text{MSE}} = \|x - \hat{x}\|^2 \quad \text{Stage 1 - "warm up"}$$

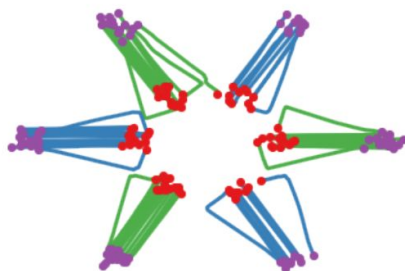
vs.

$$\mathbb{D}_{\text{LPIPS}} = \sum_l \frac{1}{H_l W_l} \|w^l \odot (\phi_l(x) - \phi_l(\hat{x}))\|^2 \quad \text{Stage 2 - "polish"}$$

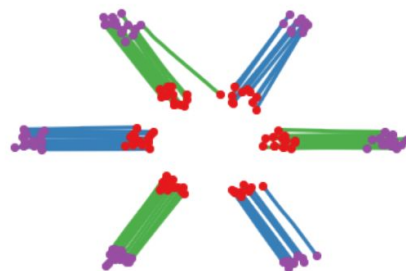
Question. Trade-off between more reflows vs. distillation?



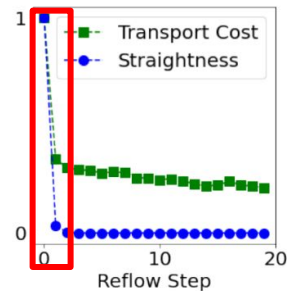
(a) The 1st rectified flow Z^1
 $Z^1 = \text{RectFlow}((X_0, X_1))$



(b) Reflow Z^2
 $Z^2 = \text{RectFlow}((Z_0^1, Z_1^1))$



(c) Reflow Z^3
 $Z^3 = \text{RectFlow}((Z_0^2, Z_1^2))$



(d) Transport cost, Straightness

most gains at the first reflow

Results. Distillation operating on straightened paths works wonders.

Method	Inf. Time	FID-5k	CLIP
SD 1.5 (25 step)[70]	0.88	20.1	0.318
2-RF (25 step)	0.88	21.5	0.315
PD-SD (1 step)[58]	0.09	37.2	0.275
2-RF (1 step)	0.09	47.0	0.271
InstaFlow-0.9B	0.09	23.4	0.304
PD-SD (2 step)[58]	0.13	26.0	0.297
PD-SD (4 step)[58]	0.21	26.4	0.300
2-RF (2 step)	0.13	31.3	0.296
InstaFlow-1.7B	0.12	22.4	0.309

Limitations. Computational burden.

- Needs teacher to pave the entire way first.
- Learned paths skipped.
- Best way to leverage information?

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Attractive alternative mindset.

Learn **global** mapping



Learn **local** mapping

Consistency models

CM = Consistency Models

Idea. Predict clean image from any point of the deterministic ODE path.

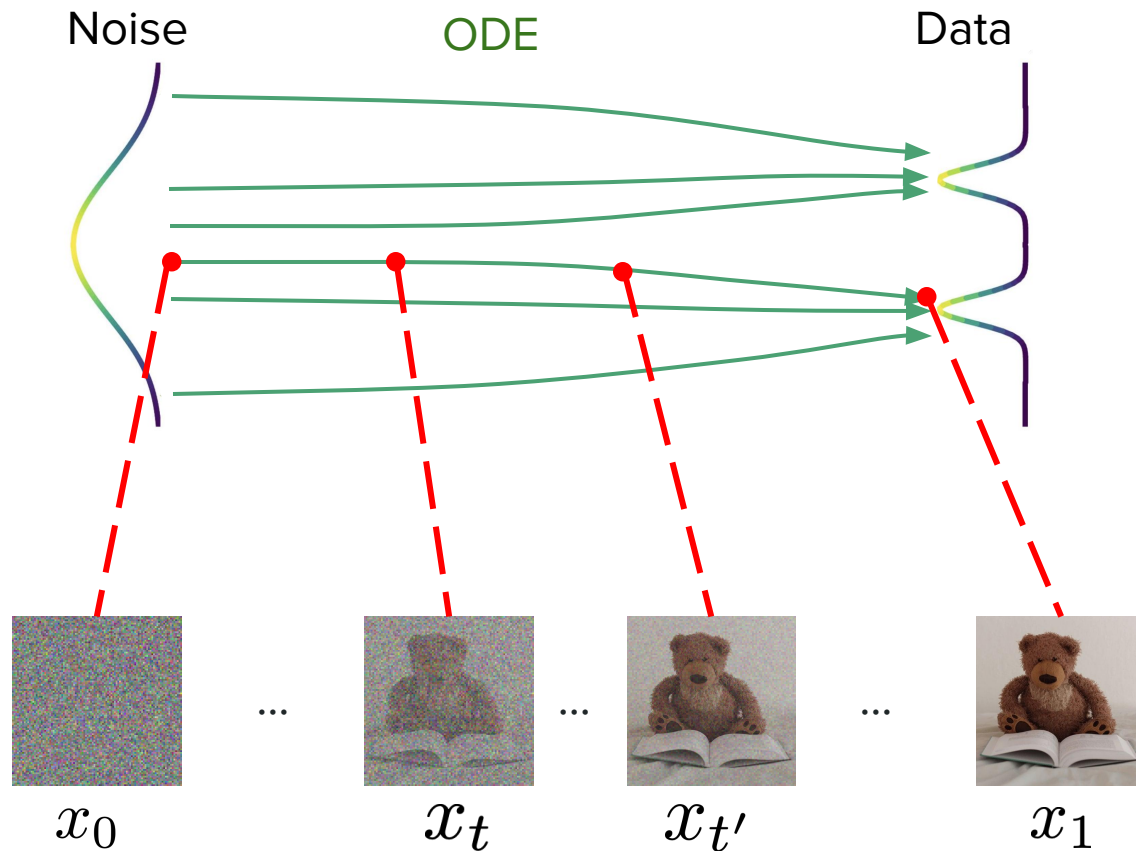
Property. Self-consistency:

$$\forall t \in [0, 1], \quad f(x_t, t) = x_1$$

Training techniques.

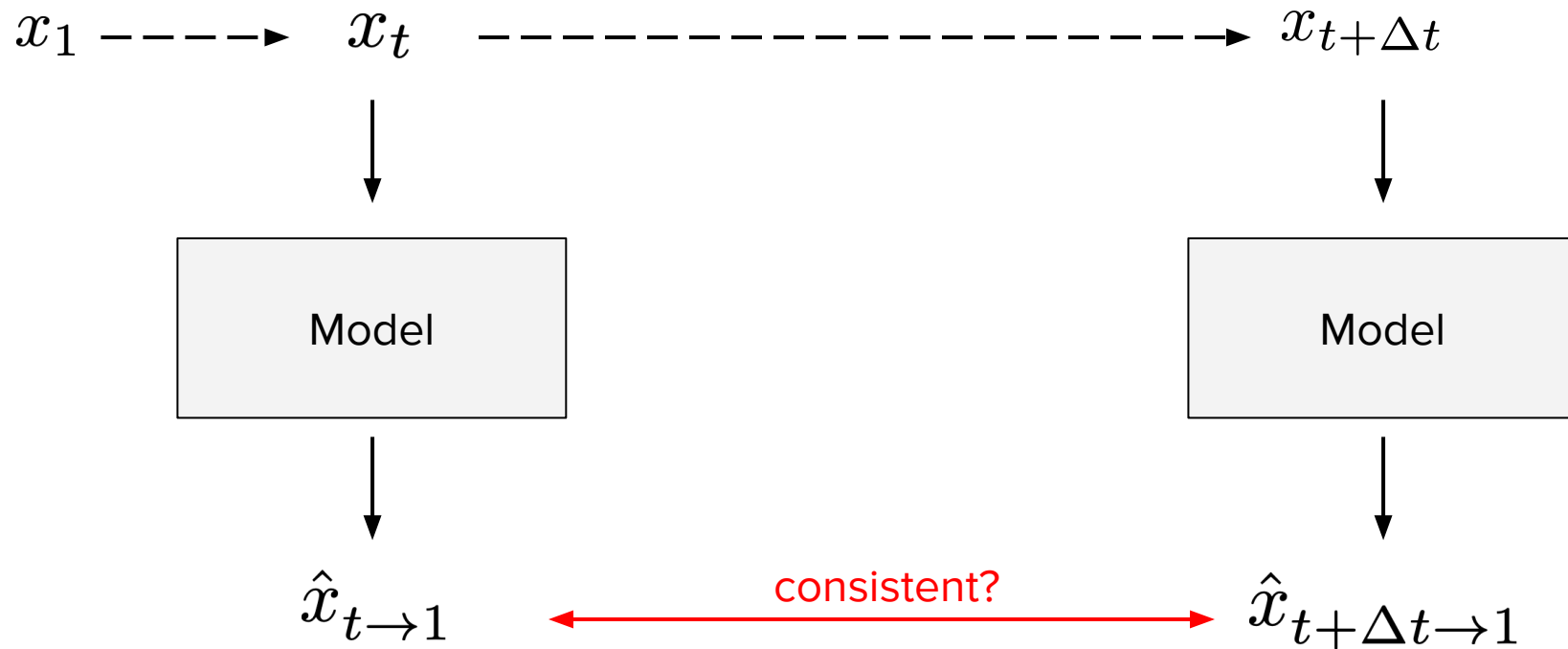
- Consistency training (CT)
- Consistency distillation (CD)

Consistency models



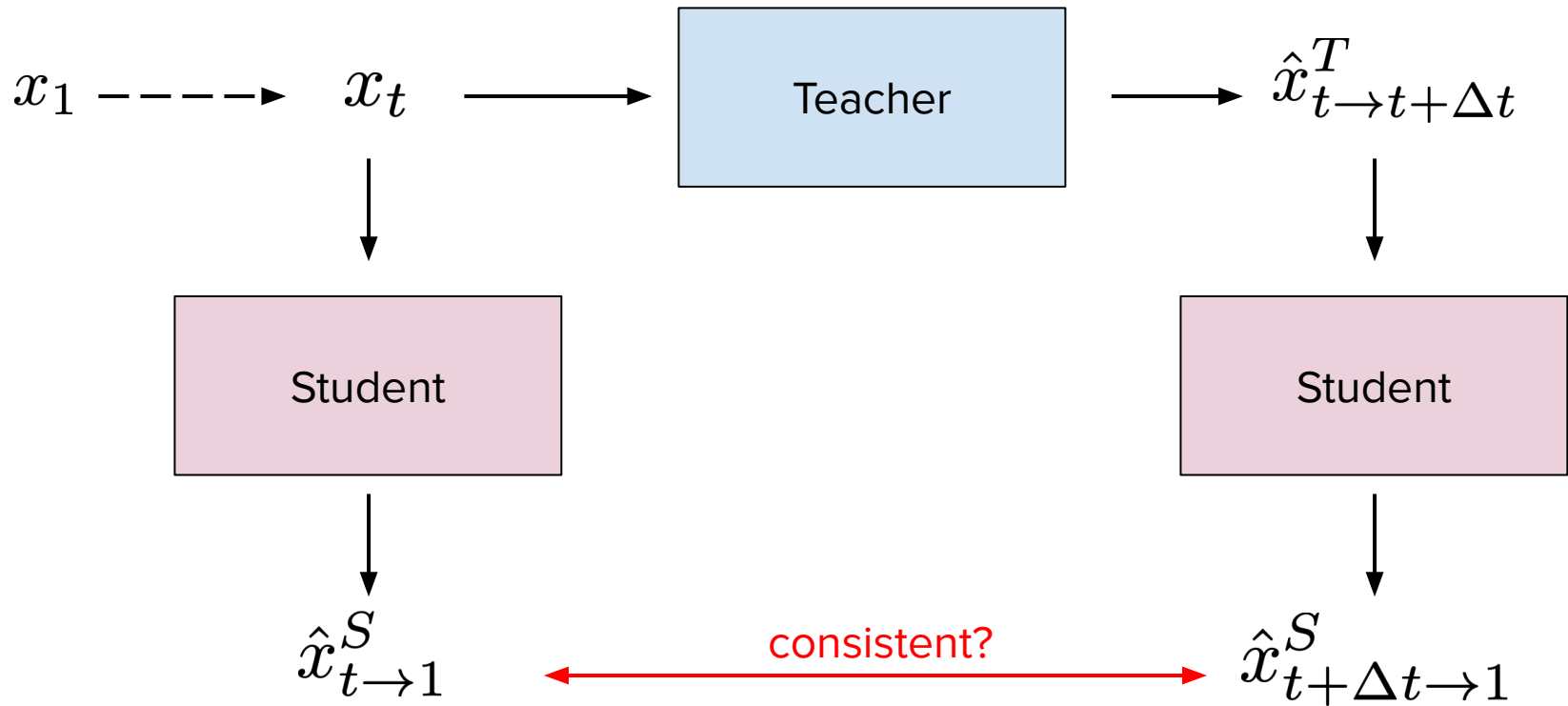
Consistency models

Consistency training.



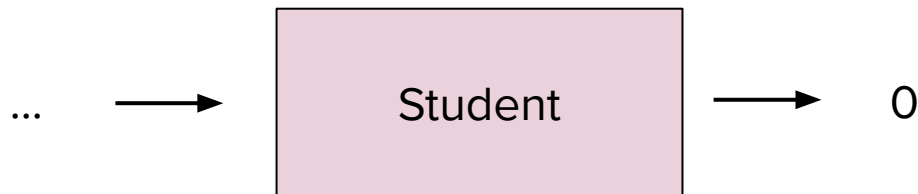
Consistency models

Consistency distillation.



Consistency models

Representation collapse?



No, thanks to boundary conditions + stop-gradient systems.

$$f_{\theta}(\mathbf{x}, t) = c_{\text{skip}}(t)\mathbf{x} + c_{\text{out}}(t)F_{\theta}(\mathbf{x}, t)$$

design choice

Consistency models

Results. Distillation wins.

METHOD	NFE (↓)	FID (↓)	IS (↑)
Diffusion + Samplers			
DDIM (Song et al., 2020)	50	4.67	
DDIM (Song et al., 2020)	20	6.84	
DDIM (Song et al., 2020)	10	8.23	
DPM-solver-2 (Lu et al., 2022)	10	5.94	
DPM-solver-fast (Lu et al., 2022)	10	4.70	
3-DEIS (Zhang & Chen, 2022)	10	4.17	
Diffusion + Distillation			
Knowledge Distillation* (Luhman & Luhman, 2021)	1	9.36	
DFNO* (Zheng et al., 2022)	1	4.12	
1-Rectified Flow (+distill)* (Liu et al., 2022)	1	6.18	9.08
2-Rectified Flow (+distill)* (Liu et al., 2022)	1	4.85	9.01
3-Rectified Flow (+distill)* (Liu et al., 2022)	1	5.21	8.79
PD (Salimans & Ho, 2022)	1	8.34	8.69
CD	1	3.55	9.48
PD (Salimans & Ho, 2022)	2	5.58	9.05
CD	2	2.93	9.75

Direct Generation

BigGAN (Brock et al., 2019)	1	14.7	9.22
Diffusion GAN (Xiao et al., 2022)	1	14.6	8.93
AutoGAN (Gong et al., 2019)	1	12.4	8.55
E2GAN (Tian et al., 2020)	1	11.3	8.51
ViTGAN (Lee et al., 2021)	1	6.66	9.30
TransGAN (Jiang et al., 2021)	1	9.26	9.05
StyleGAN2-ADA (Karras et al., 2020)	1	2.92	9.83
StyleGAN-XL (Sauer et al., 2022)	1	1.85	
Score SDE (Song et al., 2021)	2000	2.20	9.89
DDPM (Ho et al., 2020)	1000	3.17	9.46
LSGM (Vahdat et al., 2021)	147	2.10	
PFGM (Xu et al., 2022)	110	2.35	9.68
EDM (Karras et al., 2022)	35	2.04	9.84
1-Rectified Flow (Liu et al., 2022)	1	378	1.13
Glow (Kingma & Dhariwal, 2018)	1	48.9	3.92
Residual Flow (Chen et al., 2019)	1	46.4	
GLFlow (Xiao et al., 2019)	1	44.6	
DenseFlow (Grcić et al., 2021)	1	34.9	
DC-VAE (Parmar et al., 2021)	1	17.9	8.20
CT	1	8.70	8.49
CT	2	5.83	8.85

Consistency models

Limitations. Quality.

- Regression to the mean: blurriness, smoothed out results.
- Loss penalizes when exact pixels of target image are missed, even if the image is valid, realistic.

Consistency models

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- Regression to the mean: blurriness, smoothed out results.
- Loss penalizes when exact pixels of target image are missed, even if the image is valid, realistic.

Attractive alternative mindset.

Map to **sample**

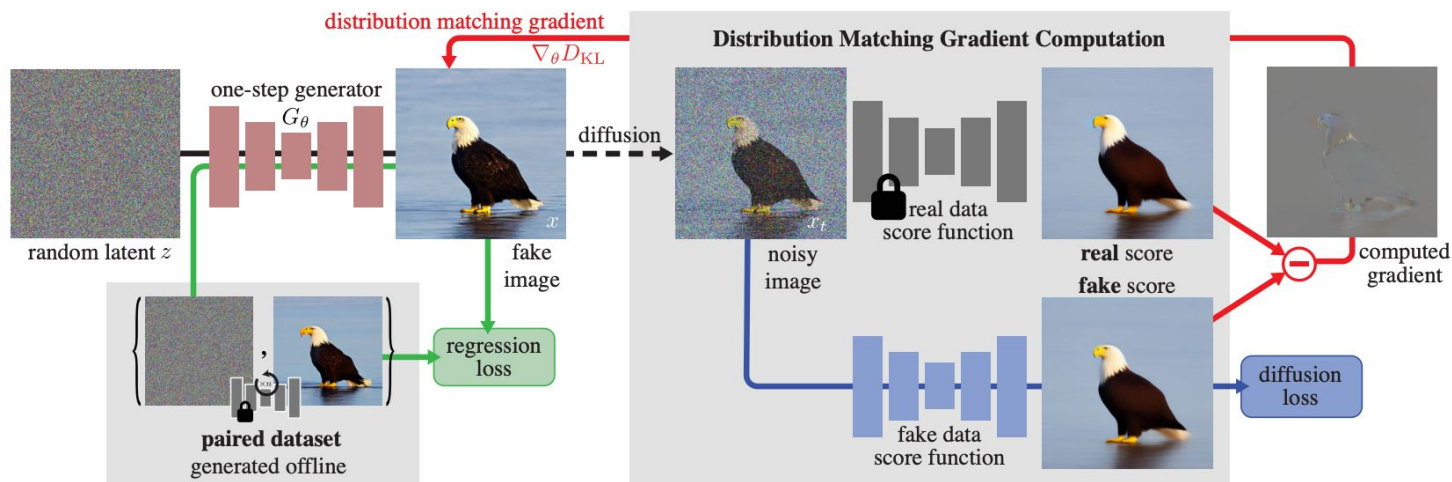


Map to **distribution**

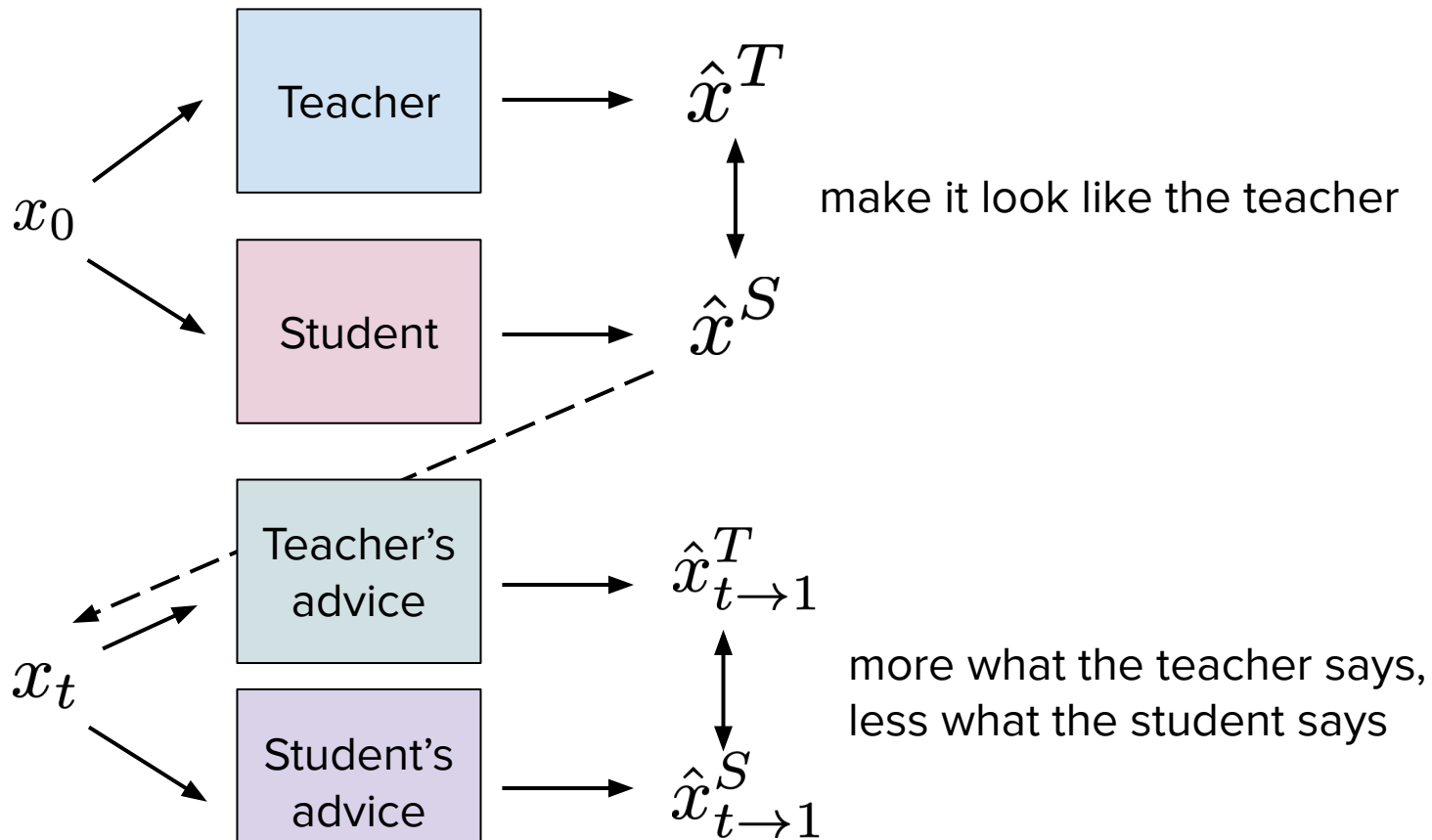
Distribution matching distillation

DMD = Distribution Matching Distillation

Idea. Match output distribution between teacher and student.



Distribution matching distillation



Distribution matching distillation

Loss function.

$$D_{KL} (p_{\text{fake}} \parallel p_{\text{real}}) = \mathbb{E}_{x \sim p_{\text{fake}}} \left(\log \left(\frac{p_{\text{fake}}(x)}{p_{\text{real}}(x)} \right) \right)$$



intractable!

Distribution matching distillation

Loss function.

$$D_{KL} (p_{\text{fake}} \parallel p_{\text{real}}) = \mathbb{E}_{x \sim p_{\text{fake}}} \left(\log \left(\frac{p_{\text{fake}}(x)}{p_{\text{real}}(x)} \right) \right)$$

Gradient update.

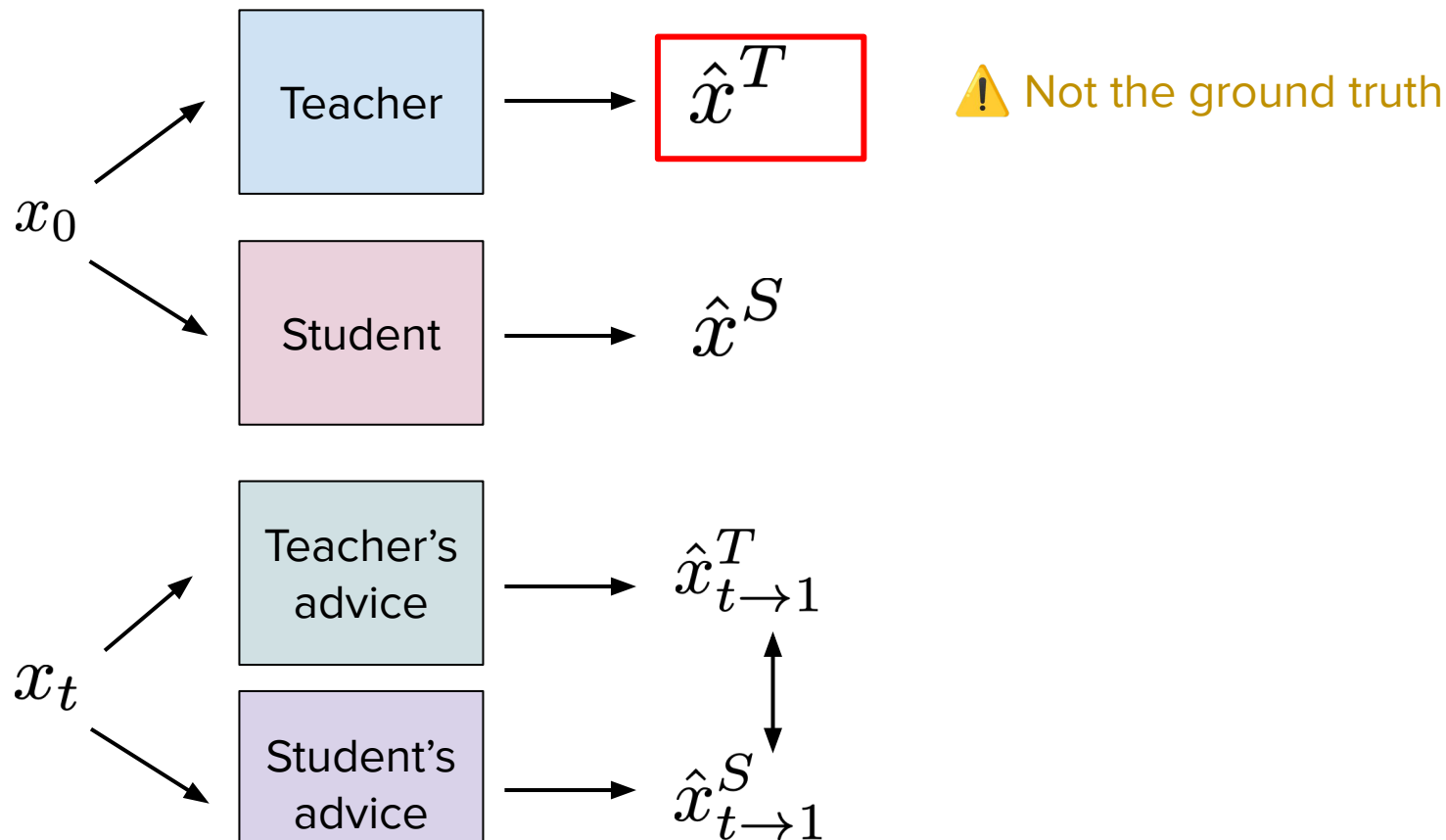
$$\nabla_{\theta} D_{KL} = \mathbb{E}_{\substack{z \sim \mathcal{N}(0; \mathbf{I}) \\ x = G_{\theta}(z)}} \left[- \left(s_{\text{real}}(x) - s_{\text{fake}}(x) \right) \frac{dG}{d\theta} \right]$$

Distribution matching distillation

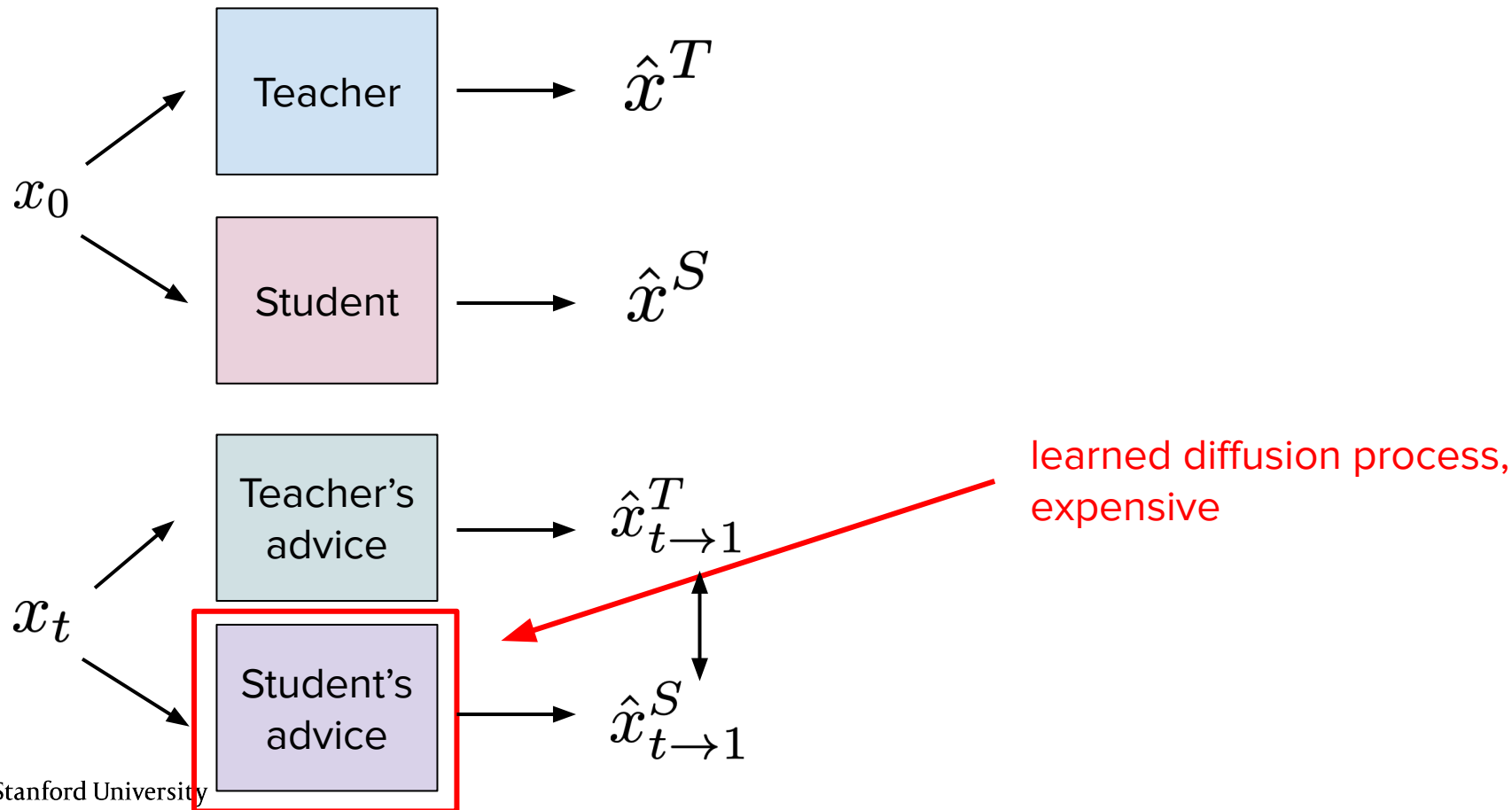
Results. Competitive.

Method	# Fwd Pass (↓)	FID (↓)
BigGAN-deep [4]	1	4.06
ADM [9]	250	2.07
Progressive Distillation [65]	1	15.39
DFNO [92]	1	7.83
BOOT [16]	1	16.30
TRACT [3]	1	7.43
Meng et al. [51]	1	7.54
Diff-Instruct [50]	1	5.57
Consistency Model [75]	1	6.20
DMD (Ours)	1	2.62
EDM [†] (Teacher) [31]	512	2.32

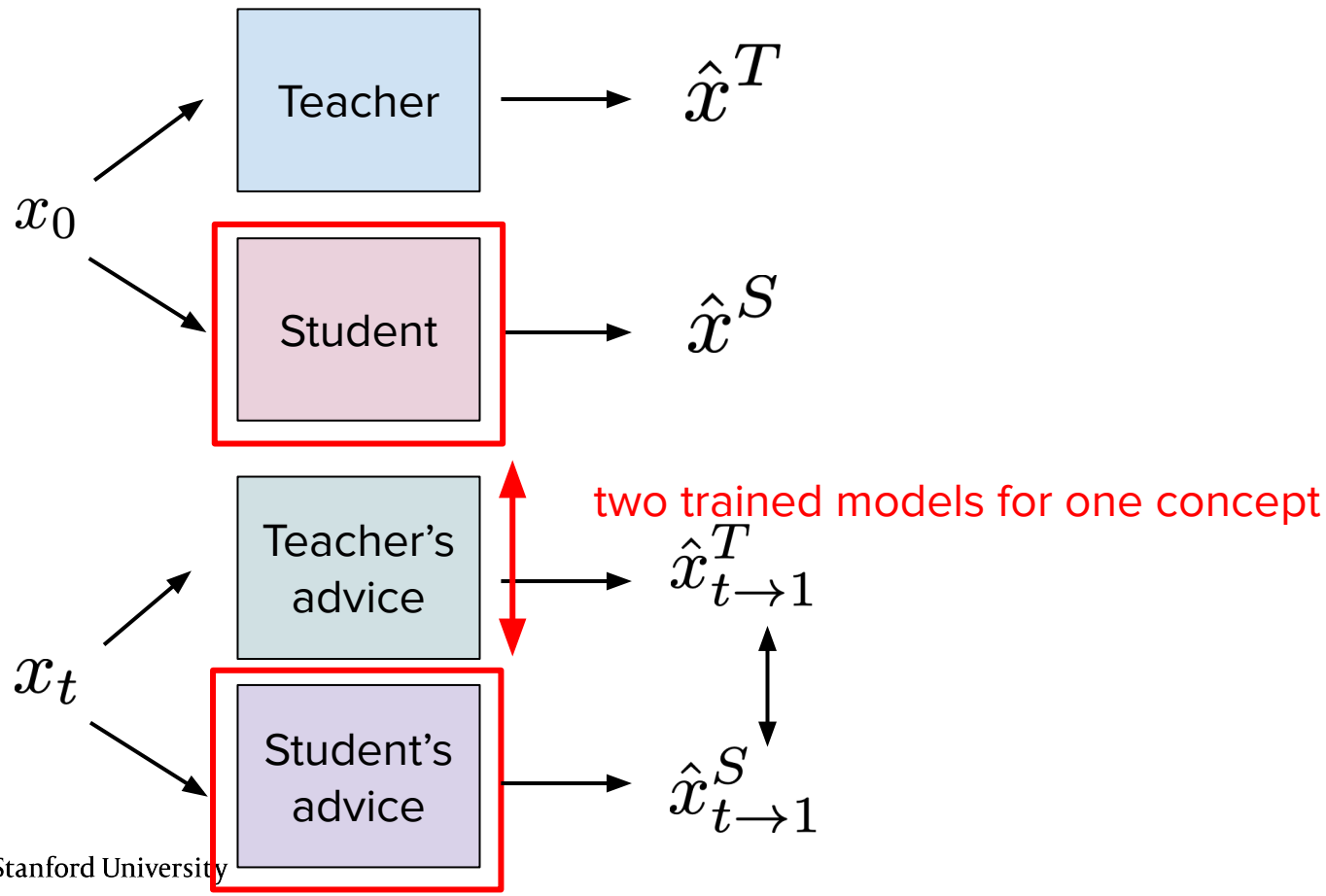
Distribution matching distillation



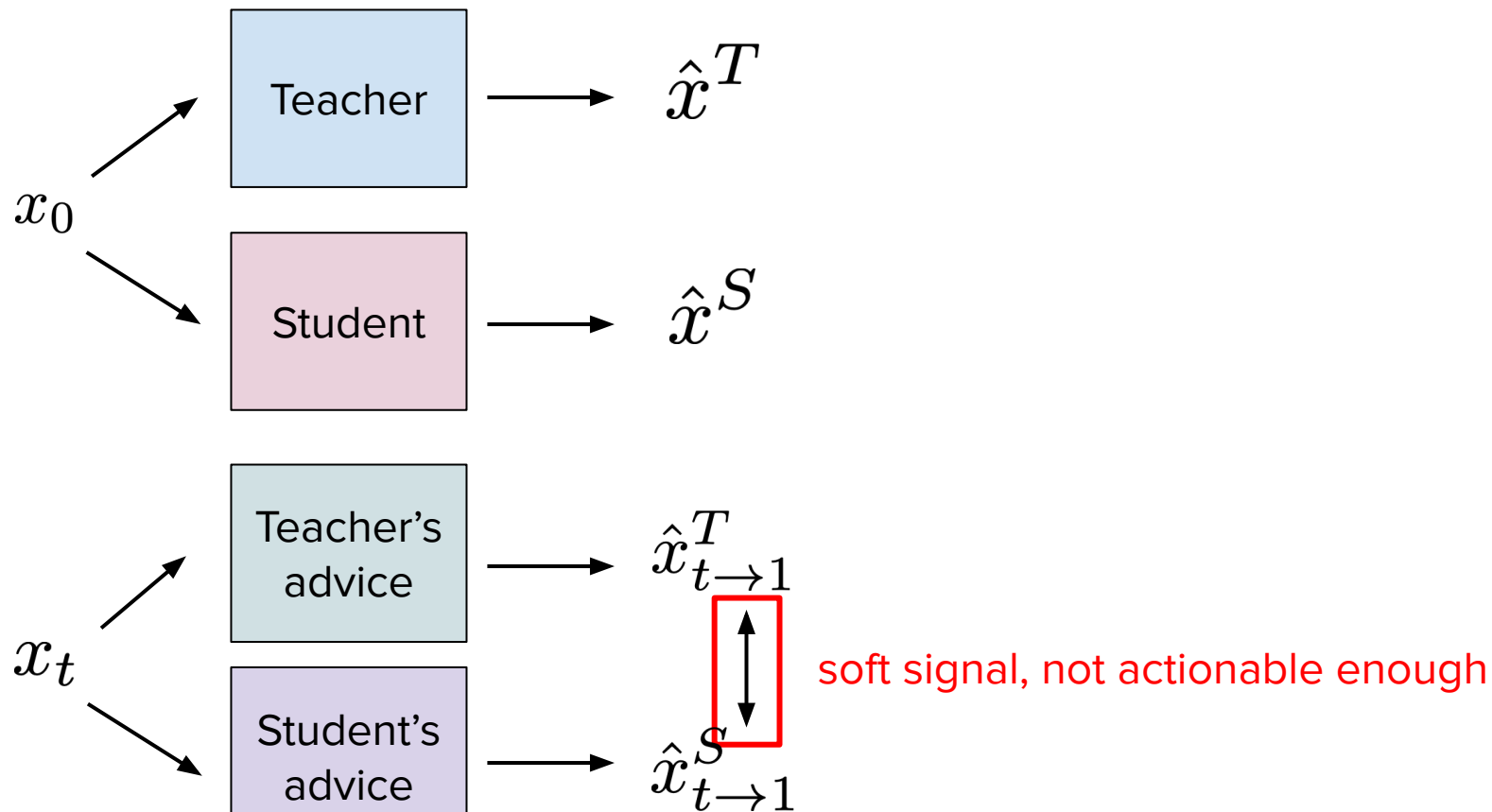
Distribution matching distillation



Distribution matching distillation



Distribution matching distillation



Distribution matching distillation

Limitations. Computational nightmare.

- Mini-diffusion training loop inside distillation loop.
- Two models compute roughly the same thing
- How to steer student with a "hard" signal?

Distribution matching distillation

Limitations. Computational nightmare.

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Attractive alternative mindset.

Fake **diffusion**

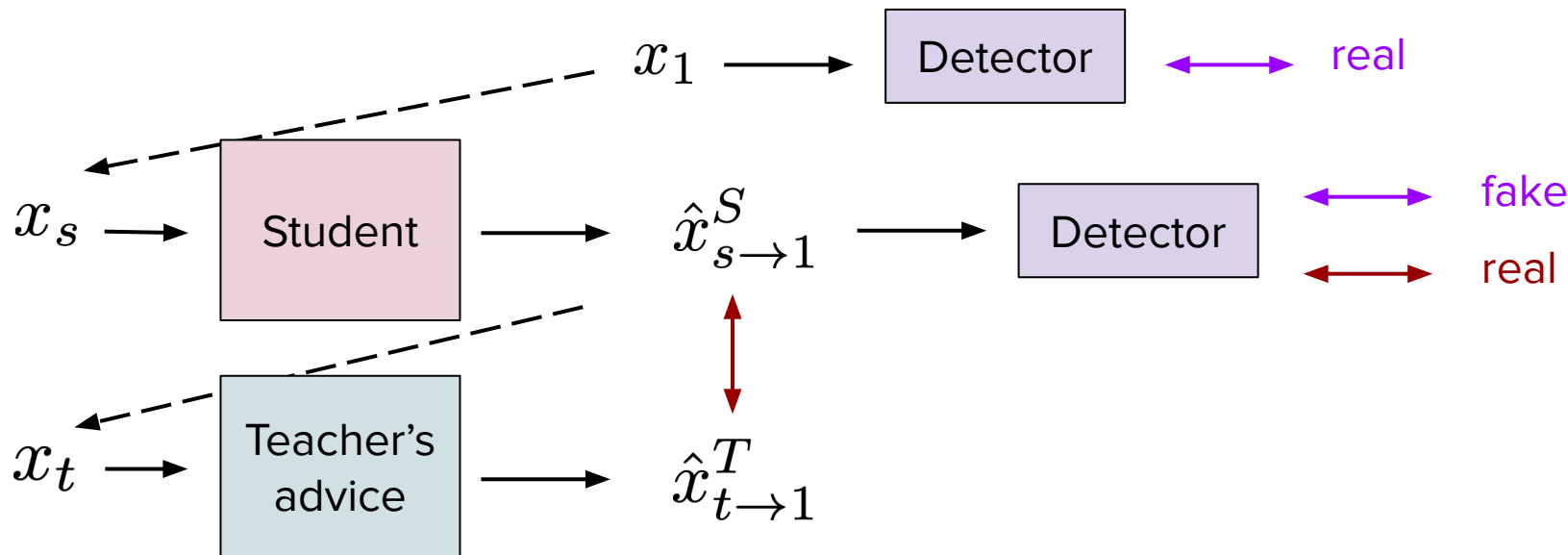


Fake **classification**

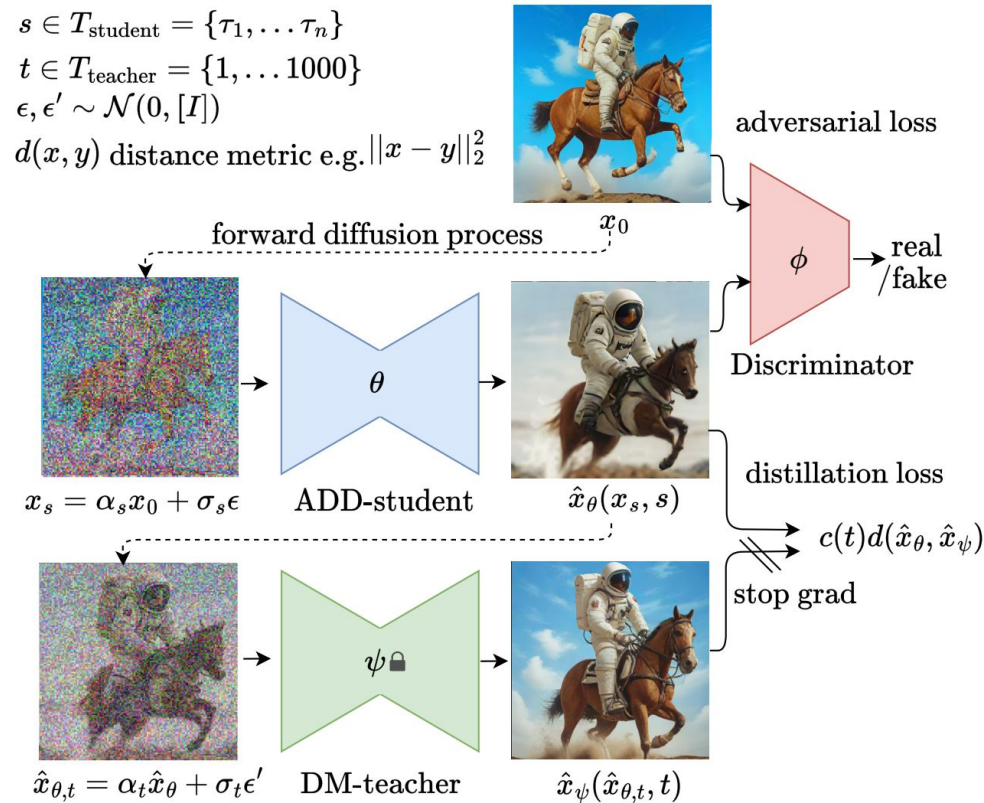
Adversarial diffusion distillation

ADD = Adversarial Diffusion Distillation

Idea. Incorporate wins from GAN by bringing in a discriminator.



Adversarial diffusion distillation



Adversarial diffusion distillation

Overall loss.

$$\mathcal{L} = \mathcal{L}_{\text{adv}}^{\text{G}}(\hat{x}_{\theta}(x_s, s), \phi) + \lambda \mathcal{L}_{\text{distill}}(\hat{x}_{\theta}(x_s, s), \psi)$$

Adversarial diffusion distillation

Overall loss.

$$\mathcal{L} = \mathcal{L}_{\text{adv}}^{\text{G}}(\hat{x}_{\theta}(x_s, s), \phi) + \lambda \mathcal{L}_{\text{distill}}(\hat{x}_{\theta}(x_s, s), \psi)$$

Discriminator incentive.

$$\begin{aligned} & \mathbb{E}_{x_0} \left[\sum_k \max(0, 1 - \mathcal{D}_{\phi,k}(F_k(x_0))) + \gamma R1(\phi) \right] \\ & + \mathbb{E}_{\hat{x}_{\theta}} \left[\sum_k \max(0, 1 + \mathcal{D}_{\phi,k}(F_k(\hat{x}_{\theta}))) \right] \end{aligned}$$

Adversarial diffusion distillation

Overall loss.

$$\mathcal{L} = \mathcal{L}_{\text{adv}}^{\text{G}}(\hat{x}_{\theta}(x_s, s), \phi) + \lambda \mathcal{L}_{\text{distill}}(\hat{x}_{\theta}(x_s, s), \psi)$$

Discriminator incentive.

$$\begin{aligned} & \mathbb{E}_{x_0} \left[\sum_k \max(0, 1 - \mathcal{D}_{\phi, k}(F_k(x_0))) + \gamma R1(\phi) \right] \\ & + \mathbb{E}_{\hat{x}_{\theta}} \left[\sum_k \max(0, 1 + \mathcal{D}_{\phi, k}(F_k(\hat{x}_{\theta}))) \right] \end{aligned}$$

Student incentive.

$$-\mathbb{E}_{s, \epsilon, x_0} \left[\sum_k \mathcal{D}_{\phi, k}(F_k(\hat{x}_{\theta}(x_s, s))) \right]$$

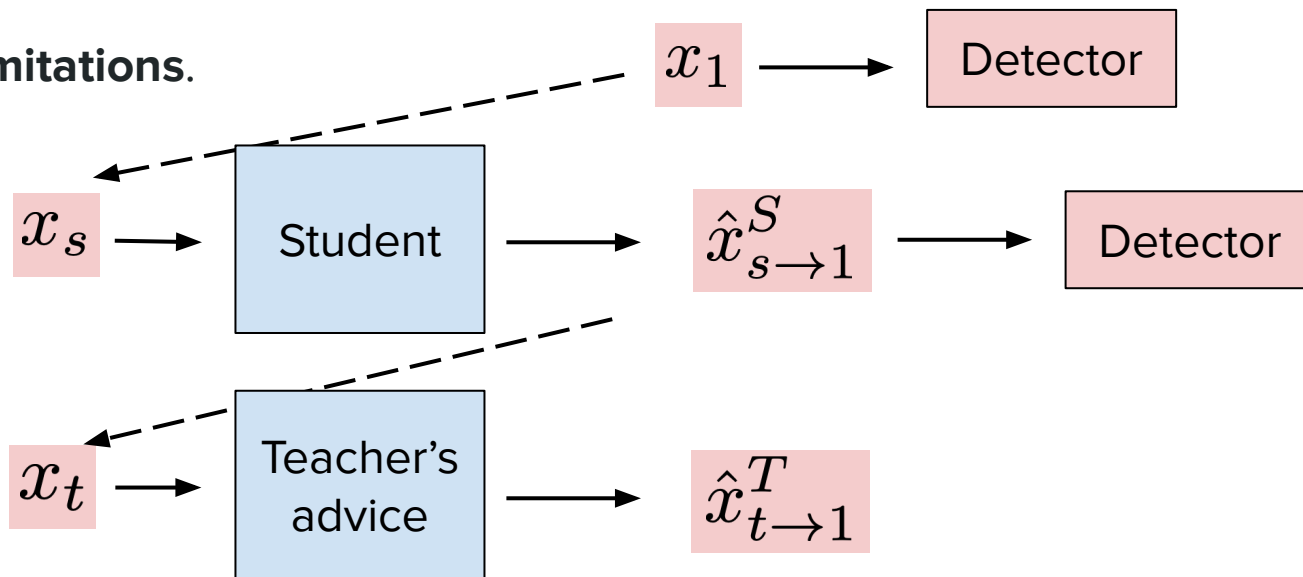
Adversarial diffusion distillation

Results.

Method	#Steps	Time (s)	FID ↓	CLIP ↑
DPM Solver [37]	25	0.88	20.1	0.318
	8	0.34	31.7	0.320
Progressive Distillation [43]	1	0.09	37.2	0.275
	2	0.13	26.0	0.297
	4	0.21	26.4	0.300
CFG-Aware Distillation [31]	8	0.34	24.2	0.300
InstaFlow-0.9B [36]	1	0.09	23.4	0.304
InstaFlow-1.7B [36]	1	0.12	22.4	0.309
UFOGen [71]	1	0.09	22.5	0.311
ADD-M	1	0.09	19.7	0.326

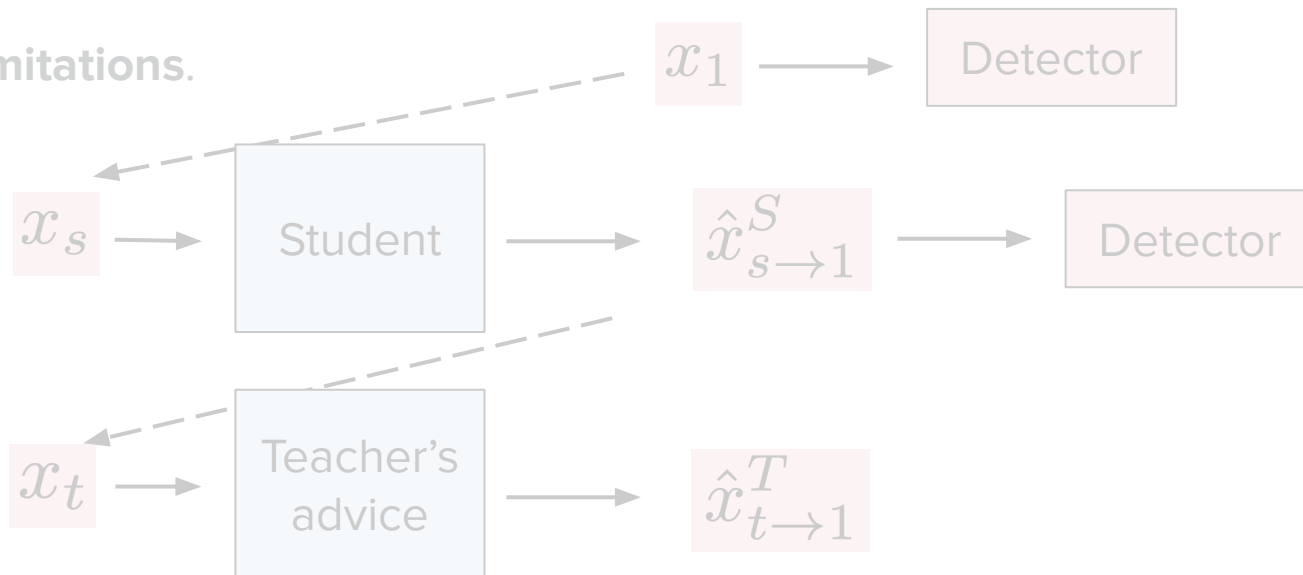
Adversarial diffusion distillation

Limitations.



Adversarial diffusion distillation

Limitations.



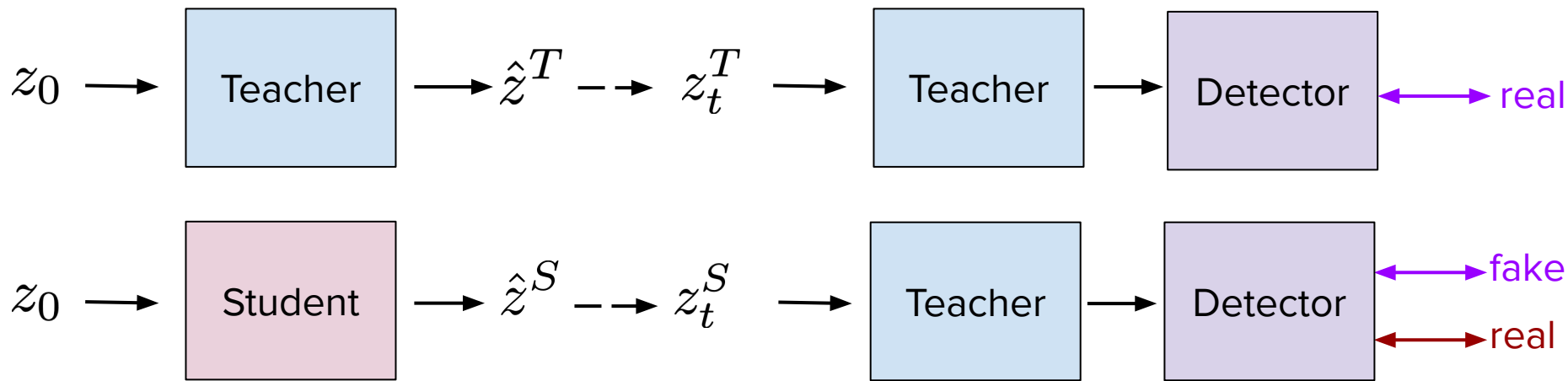
Attractive alternative mindset.

Multiple conversions $\longrightarrow$ **No conversion**

Latent adversarial diffusion distillation

LADD = Latent Adversarial Diffusion Distillation

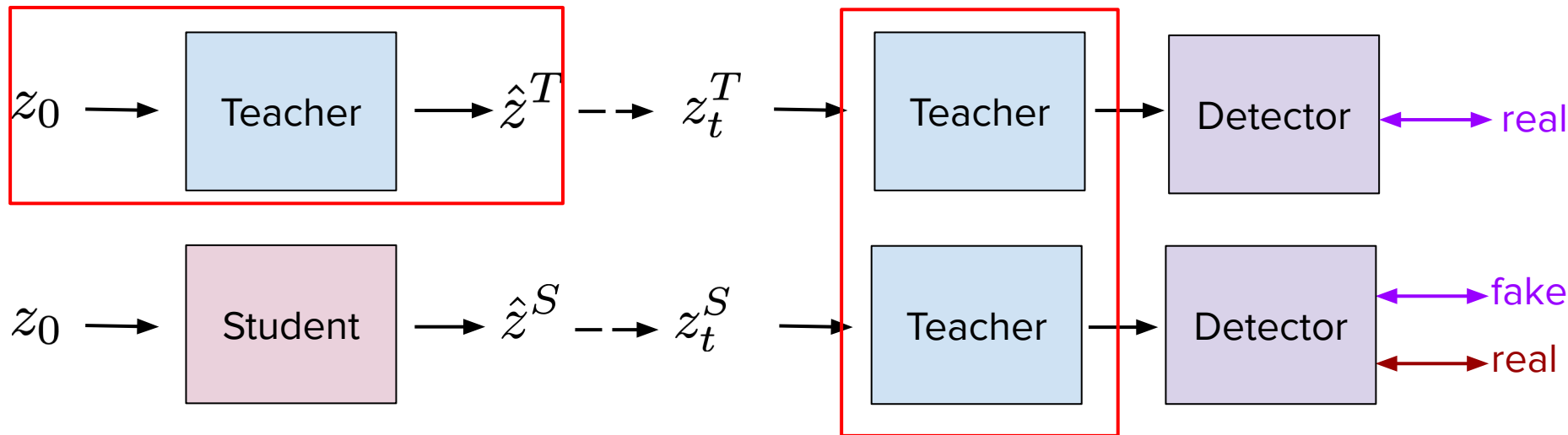
Idea. Reframe the discriminator loss in the latent space.



Latent adversarial diffusion distillation

“contains” distillation objective

feature extractor



Conclusion

Methodology.

- Teacher/student combo is powerful
- Complexity vs. quality trade-off

Conclusion

Methodology.

- Teacher/student combo is powerful
- Complexity vs. quality trade-off

Recurring ingredients.

- Regression loss: MSE/LPIPS
- Distribution/adversarial pressure
- Latent space training

Thank you for your attention!